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Global Technologies

# XC24

## The local dimming controller for current drivers

**24 Channels**

## General Description

The XC24 is a local dimming controller featuring 24 channels to drive its companion chips, XD12 LED drivers that act as the voltage-based low side current sink for White LED back-lighting solution. This controller interfaces with XD12 drivers through one serial line to send and receive the data to be needed for local dimming.

The XC24 supports the SPI interface to communicate with the host, and the SPI interface can be daisy-chained, so it is effective to expand to support a large number of dimming areas.

For raising up dimming controllability, XC24 supports the FB signal to efficiently control the LED power supply voltage and two interrupt pins, nINT_FAULT to indicate the fault events like as the thermal status, the open detection, the short detection and the feedback status, and nINT_LD to indicate the room of internal buffers for dimming data.

The XC24 can support BIST functionality to ensure Back Light Unit is properly assembled.

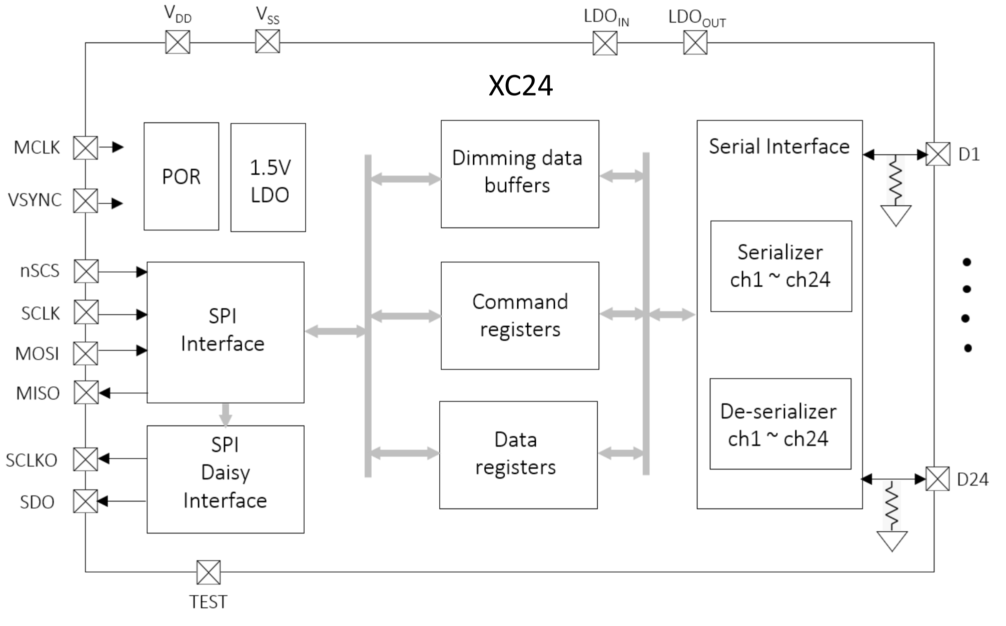
## Application

LED back-lighting with local dimming control

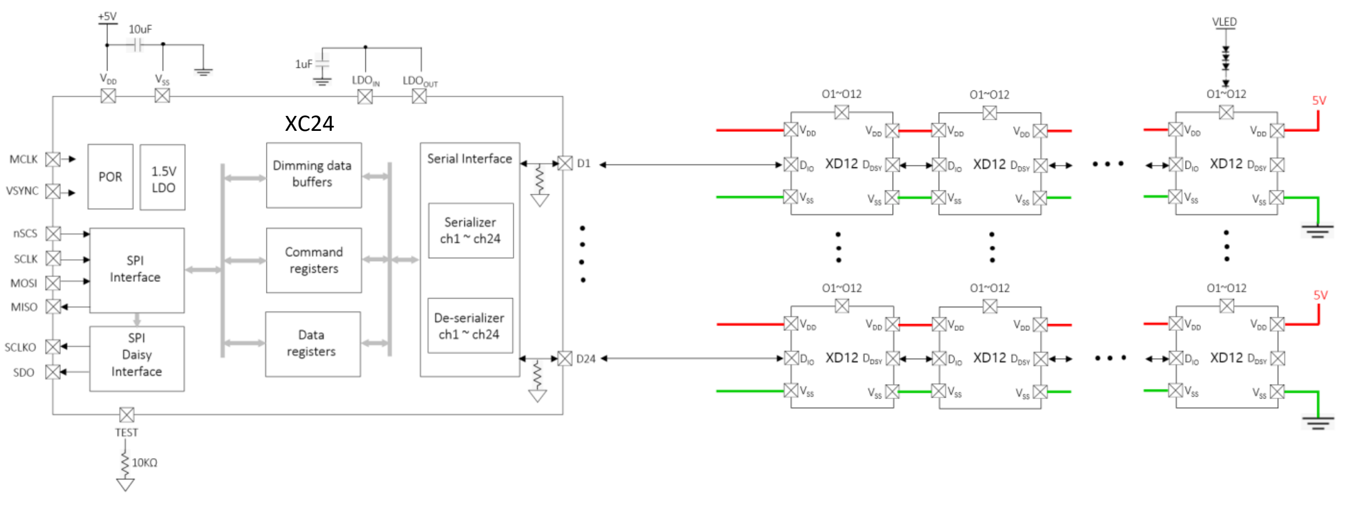
## Feature

- 24 channels local dimming controller for the XD12 LED drivers
- Flexible sizing for each channel
  - up to 31 daisied LED drivers per channel
  - up to 255 blocks per channel
  - up to 6,120 blocks per XC24
- SPI interface
  - up to 30MHz
  - support daisy-chain connection
- Configurable the baud-rate speed
- Frame synchronization support : Vsync 50~ 960Hz
- Fault events support of drivers through the interrupt function
  - Over-temperature event
  - Open detection
  - Short detection
  - FB status for VLED power efficiency
- TTL-level input range
- Supply voltage 4.5V ~ 5.5V
- 48-pin QFN 6060 Package
- Latch-up Immunity ILATCH: JEDEC JESD78D Nov 2011: ±100mA
- ESD, HBM protection (Norm: MIL 883 E Method 3015 Human Body Model): ±2kV

## Block Diagram




## Application Circuit

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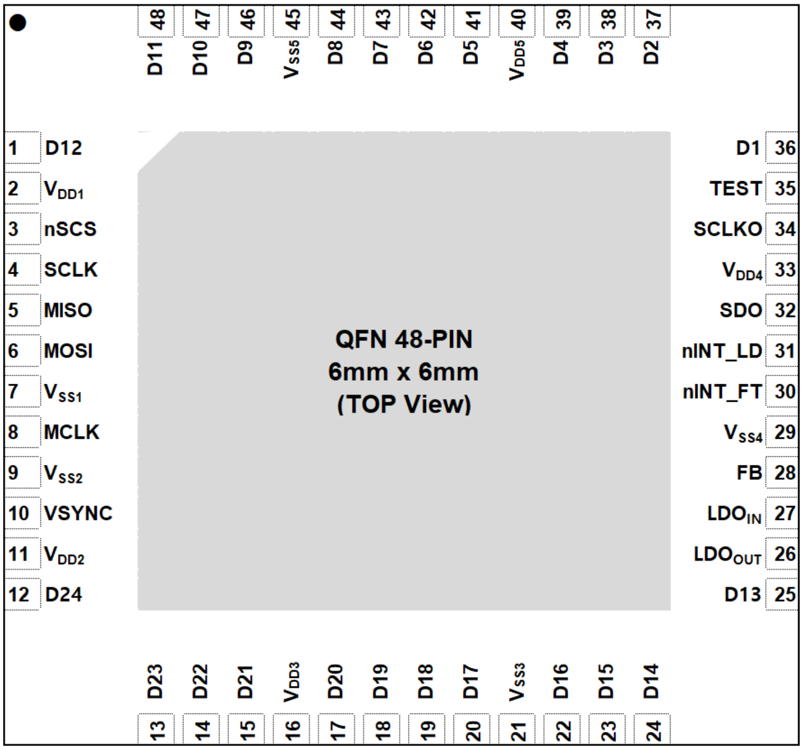
## Pin diagram (Top View)




## Pin description

| Pin Name | Pin No. | I/O | Type | Description | Remark |
|---|---|---|---|---|---|
| D12 | 1 | I/O | Digital | Data in/out | |
| $V_{DD1}$ | 2 | I | Power | Digital $V_{DD}$ | |
| nSCS | 3 | I | Digital | SPI Chip Select | |
| SCLK | 4 | I | Digital | SPI SCLK | |
| MISO | 5 | O | Digital | SPI MISO | |
| MOSI | 6 | I | Digital | SPI MOSI | |
| $V_{SS1}$ | 7 | I | GND | Digital GND | |
| MCLK | 8 | I | Digital | Master Clock | |

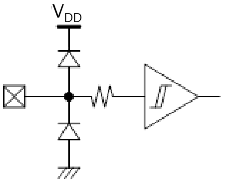
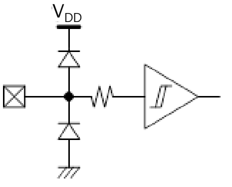
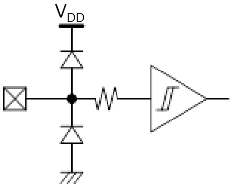
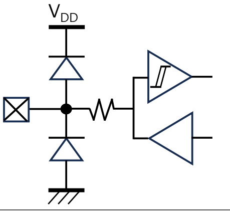
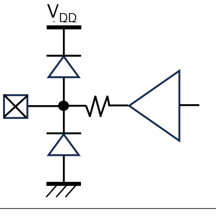
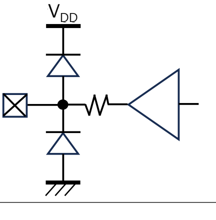
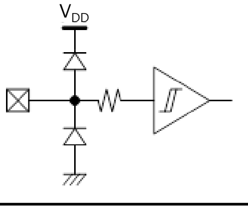
## Pin description

| Pin Name | Pin No. | I/O | Type | Description | Remark |
|---|---|---|---|---|---|
| $V_{SS2}$ | 9 | I | GND | Digital GND | |
| VSYNC | 10 | I | Digital | Frame synchronization | |
| $V_{DD2}$ | 11 | I | Power | Digital $V_{DD}$ | |
| D24 | 12 | I/O | Digital | Data in/out | |
| D23 | 13 | I/O | Digital | Data in/out | |
| D22 | 14 | I/O | Digital | Data in/out | |
| D21 | 15 | I/O | Digital | Data in/out | |
| $V_{DD3}$ | 16 | I | Power | Digital $V_{DD}$ | |
| D20 | 17 | I/O | Digital | Data in/out | |
| D19 | 18 | I/O | Digital | Data in/out | |
| D18 | 19 | I/O | Digital | Data in/out | |
| D17 | 20 | I/O | Digital | Data in/out | |
| $V_{SS3}$ | 21 | I | GND | Digital GND | |
| D16 | 22 | I/O | Digital | Data in/out | |
| D15 | 23 | I/O | Digital | Data in/out | |
| D14 | 24 | I/O | Digital | Data in/out | |
| D13 | 25 | I/O | Digital | Data in/out | |
| $LDO_{IN}$ | 26 | I | Power | LDO feedback from $LDO_{OUT}$ | Attach 1uF |
| $LDO_{OUT}$ | 27 | O | Power | Internal LDO out to $LDO_{IN}$ | Connect to $LDO_{IN}$ |
| FB | 28 | O | Digital | Feedback for LED power supply | Open-drain, 10kΩ PU |
| $V_{SS4}$ | 29 | I | GND | Digital GND | |
| nINT_FT | 30 | O | Digital | Interrupt for fault events | Open-drain, 10kΩ PU |
| nINT_LD | 31 | O | Digital | Interrupt for dimming data room | Open-drain, 10kΩ PU |
| SDO | 32 | O | Digital | SPI MOSI daisied out | Unused : 10kΩ PD |
| $V_{DD4}$ | 33 | I | Power | Digital $V_{DD}$ | |
| SCLKO | 34 | O | Digital | SPI SCLK daisied out | Unused : 10kΩ PD |

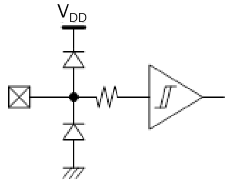
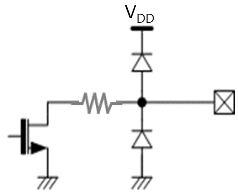
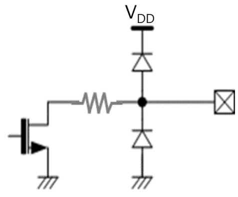
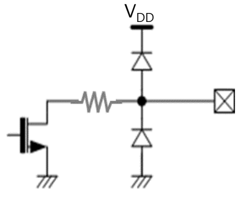
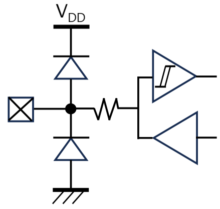
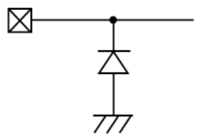
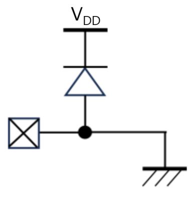
## Pin description

| Pin Name | Pin No. | I/O | Type | Description | Remark |
|---|---|---|---|---|---|
| TEST | 35 | I | Digital | TEST mode pin | 10Ω PD |
| D1 | 36 | I/O | Digital | Data in/out | |
| D2 | 37 | I/O | Digital | Data in/out | |
| D3 | 38 | I/O | Digital | Data in/out | |
| D4 | 39 | I/O | Digital | Data in/out | |
| $V_{DD5}$ | 40 | I | Power | Digital $V_{DD}$ | |
| D5 | 41 | I/O | Digital | Data in/out | |
| D6 | 42 | I/O | Digital | Data in/out | |
| D7 | 43 | I/O | Digital | Data in/out | |
| D8 | 44 | I/O | Digital | Data in/out | |
| $V_{SS5}$ | 45 | I | GND | Digital GND | |
| D9 | 46 | I/O | Digital | Data in/out | |
| D10 | 47 | I/O | Digital | Data in/out | |
| D11 | 48 | I/O | Digital | Data in/out | |

## Equivalent Circuits of individual pins

| Pad Name | I/O | Internal Circuit | Description |
|---|---|---|---|
| nSCS | I |  | SPI Chip Select |
| SCLK | I |  | SPI Clock Input |
| MOSI | I |  | SPI MOSI |
| MISO | O |  | SPI MISO |
| SCLKO | O |  | SCLK Daisy Out |
| SDO | O |  | MOSI Daisy Out |
| MCLK | I |  | Mater Clock |

## Equivalent Circuits of individual pins

| Pad Name | I/O | Internal Circuit | Description |
|---|---|---|---|
| VSYNC | I |  | Frame Sync. |
| nINT_FB | O |  | Interrupt for Fault |
| nINT_LD | O |  | Interrupt for LD |
| FB | O |  | Feedback for VLED |
| D | I/O |  | Data In/Out |
| $V_{DD}$ | - |  | Digital Power |
| $V_{SS}$ | - |  | Digital GND |

## Maximum Absolute ratings (Ta = 25℃ unless otherwise noted)

| Parameter | Symbol | Value | Unit |
|---|---|---|---|
| Supply voltage | $V_{DD}$ | -0.3 ~ 6.0 | V |
| MCLK to GND voltage | $V_{MCLK}$ | -0.3 ~ $V_{DD}$+0.5 | V |
| VSYNC to GND voltage | $V_{VSYNC}$ | -0.3 ~ $V_{DD}$+0.5 | V |
| nSCS to GND voltage | $V_{SCS}$ | -0.3 ~ $V_{DD}$+0.5 | V |
| SCLK to GND voltage | $V_{SCLK}$ | -0.3 ~ $V_{DD}$+0.5 | V |
| MOSI to GND voltage | $V_{MOSI}$ | -0.3 ~ $V_{DD}$+0.5 | V |
| MISO to GND voltage | $V_{MISO}$ | -0.3 ~ $V_{DD}$+0.5 | V |
| SCLKO to GND voltage | $V_{SCLK}$ | -0.3 ~ $V_{DD}$+0.5 | V |
| SDO to GND voltage | $V_{SDO}$ | -0.3 ~ $V_{DD}$+0.5 | V |
| D voltage | $V_{DIO}$ | -0.3 ~ $V_{DD}$+0.5 | V |
| Ambient temperature | $T_A$ | -20 ~ 85 | ℃ |
| Junction temperature | $T_J$ | -20 ~ 125 | ℃ |
| Storage temperature | $T_{STG}$ | -45 ~ 150 | ℃ |
| Relative Humidity(non-condensing) | $RH_{NC}$ | 5 ~ 85 | % |
| Moisture Sensitivity | MSL | Maximum Floor Life-time | 3 |

Notes : Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device

## Electrostatic Discharge

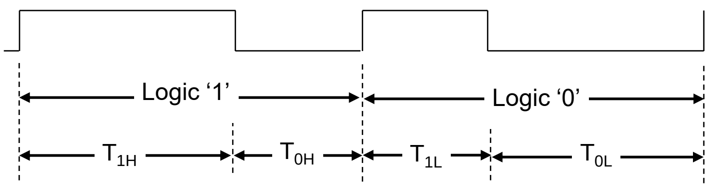
| Parameter | Symbol | Value | Unit | Remark |
|---|---|---|---|---|
| Electrostatic Discharge HBM ; Contact | ESDHBM | ±2K | V | C=100 pF, R=1.5 kΩ |
| Electrostatic Discharge CDM ; Contact | ESDCDM | ±500 | V | R=1 Ω |

## Electrical Characteristic ($V_{DD}$ =5.0V, Ta=25.0℃ unless otherwise noted)

| Characteristics | Symbol | Condition | Value | | | Unit |
|---|---|---|---|---|---|---|
| | | | min | typ | max | |
| Supply | | | | | | |
| Operating voltage | $V_{DD}$ | | 4.5 | - | 5.5 | V |
| Operating Current | $I_{DD}$ | | - | - | | mA |
| Quiescent current | $I_{DDD}$ | | - | | - | mA |
| MCLK | | | | | | |
| Frequency | $F_{MCLK}$ | | - | 14.7456 | - | MHz |
| Duty | | | 45 | 50 | 55 | % |
| Jitter | $\triangle F_{MCLK}$ | | -1 | | 1 | % |
| Input | | | | | | |
| nSCS Input Voltage | $V_{SCSL}$ | Chip Select, low level | - | - | 0.8 | V |
| | $V_{SCSH}$ | Chip Select, high level | 2.0 | - | $V_{DD}$ | V |
| SCLK Input Voltage | $V_{SCLKL}$ | SPI Clock, low level | - | - | 0.8 | V |
| | $V_{SCLKH}$ | SPI Clock, high level | 2.0 | - | $V_{DD}$ | V |
| MOSI Input Voltage | $V_{MOSIL}$ | MOSI, low level | - | - | 0.8 | V |
| | $V_{MOSIH}$ | MOSI, high level | 2.0 | - | $V_{DD}$ | V |
| VSYNC Input Voltage | $V_{VSYNCL}$ | VSYNC low level | - | - | 0.8 | V |
| | $V_{VSYNCH}$ | VSYNC high level | 2.0 | - | $V_{DD}$ | V |
| Output | | | | | | |
| MISO Output voltage | $V_{MISOL}$ | MISO, low level | - | - | 0.8 | V |
| | $V_{MISOH}$ | MISO, high level | 2.0 | - | $V_{DD}$ | V |
| SCLKO Output voltage | $V_{SCLKOL}$ | SCLKO, low level | - | - | 0.8 | V |
| | $V_{SCLKOH}$ | SCLKO, high level | 2.0 | - | $V_{DD}$ | V |
| SDO Output voltage | $V_{SDOL}$ | SDO, low level | - | - | 0.8 | V |
| | $V_{SDOH}$ | SDO, high level | 2.0 | - | $V_{DD}$ | V |
| Bi-direction | | | | | | |
| $D_{IO}$ | $V_{IL}$ | Input logic low level | - | - | 0.5 | V |
| | $V_{IH}$ | Input logic high level | 3.5 | - | $V_{DD}$ | V |

## Electrical Characteristic ($V_{DD}$ =5.0V, Ta=25.0℃ unless otherwise noted)

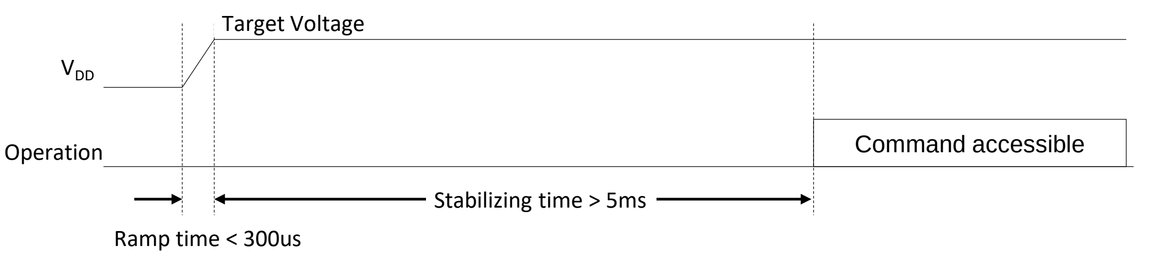
| Characteristics | Symbol | Condition | Value | | | Unit |
|---|---|---|---|---|---|---|
| | | | min | typ | max | |
| Bi-direction | | | | | | |
| $D_{IO}$ | $V_{OL}$ $V_{OH}$ | Output logic low level | - | - | 0.5 | V |
| | | Output logic high level | $V_{DD}$ - 0.5 | - | $V_{DD}$ | |
| | $R_{PD}$ | Internal pull-down resistance | - | 48 | - | KΩ |
| Logic high time 1) | $T_{1H}$, $T_{0L}$ | | - | 8 | 20 | MCLK |
| Logic low time 1) | $T_{1L}$, $T_{0H}$ | | - | 4 | 10 | MCLK |




[ Figure. 1]

Note : 1) $T_{1H} \geq T_{0H} * 2$, $T_{0L} \geq T_{1L} * 2$, $(T_{1H} + T_{0H}) = (T_{0L} + T_{1L}) \leq 31$

## Power Sequence

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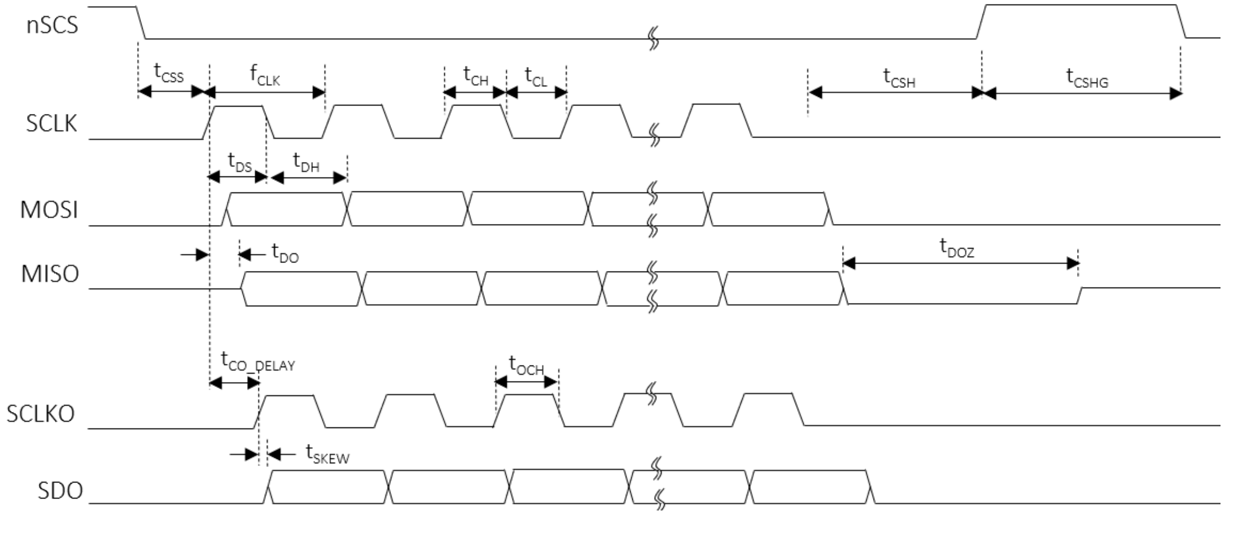
## AC Characteristics ($V_{DD\_S}$=5.0V, Ta=25.0℃ unless otherwise noted)

| Characteristics | Symbol | Condition | Value | | | Unit |
|---|---|---|---|---|---|---|
| | | | min | typ | max | |
| SCLK | | | | | | |
| SCLK frequency | $f_{CLK}$ | | - | - | 30 | MHz |
| SCLK duty | $t_{CLK\_DUTY}$ | | 48 | - | 52 | % |
| SCLK high time | $t_{CH}$ | | 8 | - | - | ns |
| SCLK low time | $t_{CL}$ | | 8 | - | - | ns |
| SCLKO | | | | | | |
| SCLK to SCLKO delay | $T_{CO\_DELAY}$ | | - | - | 4.2 (3) | ns |
| The skew between SCLKO to SDO | $t_{SKEW}$ | | - | - | 3 (3) | ns |
| SCLKO high time | $t_{OCH}$ | | $t_{CH}$ - 1(3) | - | - | ns |
| nSCS setup to the first rising SCLK | $t_{CSS}$ | | 50 | - | - | ns |
| nSCS hold time from the last falling SCLK | $t_{CSH}$ | | 3 | - | - | MCLK |
| nSCS high latency time | $t_{CSHG}$ | | 3 | - | - | MCLK |
| MOSI setup time | $t_{DS}$ | | 5 | - | - | ns |
| MOSI hold time | $t_{DH}$ | | 2 | - | - | ns |
| MISO delay from the rising edge SCLK | $t_{DO}$ | | - | - | 4.4 | ns |
| MISO high-Z from the nSCS de-assertion | $t_{DOZ}$ | | - | - | 1 | MCLK |

Note (3) It should be careful that the $t_{CO\_DELAY}$, $t_{SKEW}$, $t_{OCH}$ and $t_{DO}$ are accumulated by the number of daisy devices if the daisy chain connection is used. So, for the read and write transaction, the proper timing should be considered in daisy chain.



# Functional Description

## 1. SPI Interface

For the local dimming data transmit a Serial Peripheral Interface is used. The SPI is configurated to work as SPI slave. A Daisy Chain structure can be used to support large scale blocks array with several devices and operated up to 30MHz.

## SPI Daisy Chain Structure

All SPI slaves share the Chip select signal, and these devices in Daisy chain can be treated as one big shift register. The devices can be enumerated as described below.

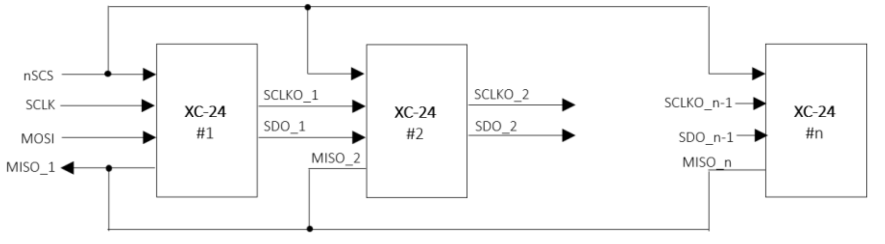



Figure 1. SPI daisy connection

## SPI Protocol

The XC24 has 3 types of the SPI Interface protocol, the register-read, the register-write and the LD data-transfer. The SPI protocol's format consists of a 16-bit command and a 16-bit register data or a series of 24(or 16)-bit LD data following the 16-bit command.

[Command format]

| | Bit15 | Bit14 | Bit13 | Bit12 | Bit11 | Bit10 | Bit9 | Bit8 | Bit7 | Bit6 | Bit5 | Bit4 | Bit3 | Bit2 | Bit1 | Bit0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Command code [1:0] | | Address [7:0] | | | | | | | | Burst size [5:0] | | | | | |
| Register-read | 0 | 1 | Start address [7:0] | | | | | | | | Burst size [5:0] | | | | | |
| Register-write | 1 | 0 | Start address [7:0] | | | | | | | | Burst size [5:0] | | | | | |
| LD-transfer | 1 | 1 | don't care | | | | | | | | Line size [5:0] | | | | | |

[Data format]

| | Bit15 | Bit14 | Bit13 | Bit12 | Bit11 | Bit10 | Bit9 | Bit8 | Bit7 | Bit6 | Bit5 | Bit4 | Bit3 | Bit2 | Bit1 | Bit0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Data [15:0] | | | | | | | | | | | | | | | |
| Register-RW | Register data [15:0] | | | | | | | | | | | | | | | |
| LD1 transfer | LD_WIDTH [1:0] = 2'b00 : 24 bits /block | | | | | | | | | | | | | | | |
| LD2 transfer | LD_WIDTH [1:0] = 2'b01 : 24 bits /block | | | | | | | | | | | | | | | |
| LD3 transfer | LD_WIDTH [1:0] = 2'b10 : 24 bits /block | | | | | | | | | | | | | | | |
| LD4 transfer | LD_WIDTH [1:0] = 2'b11 : 16 bits /block | | | | | | | | | | | | | | | |

Figure 2. SPI protocol format

# Functional Description

## SPI read/write sequence in burst

The SPI access sequence is as follows, depending on the type of access. There is no difference between a single and a burst access except for the number of data due to the burst size.

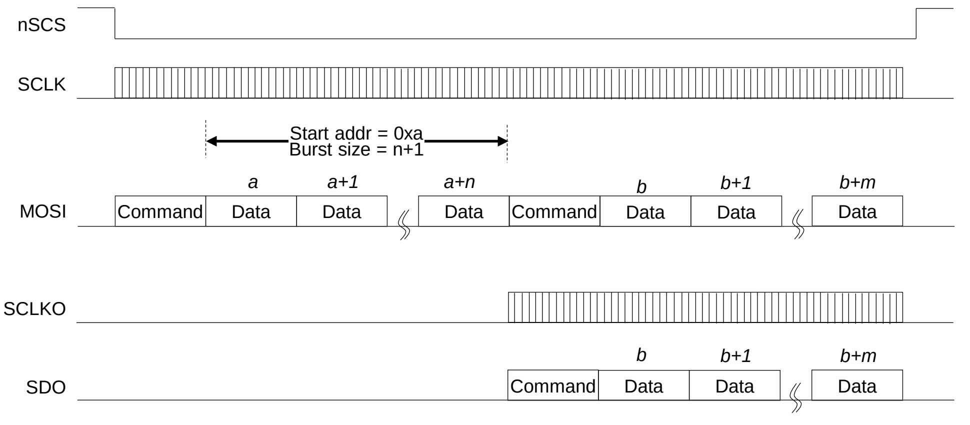



Figure 3. The SPI Burst write in daisied 2 devices

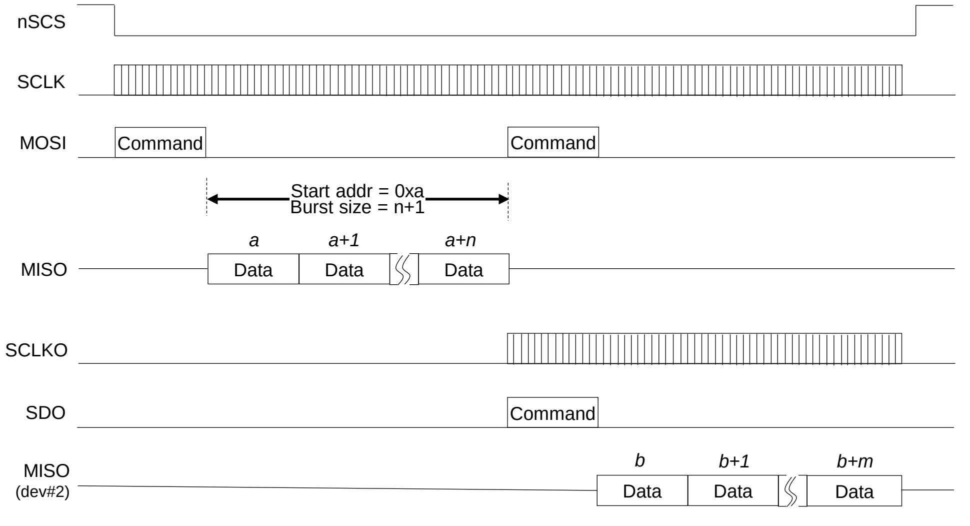



Figure 4. The SPI Burst read in daisied 2 devices

# Functional Description

## SPI LD transfer sequence

The SPI sequence for the LD transfer is as follows. The LD width depends on the LD_WIDTH value of the CHANNEL ENABLE 2 register. If the LD_WIDTH = 2'b00, then the LD data size of one block zone transferred through the SPI becomes 24 bits consisted of PAM 12 bits and PWM 12 bits. Even if the transmitted width is 24 bits, the valid bits are actually 20 bits, PAM 10 bits + PWM 10 bits. Therefore, the data size per block to be transmitted to the daisied devices is 20 bits.

| LD_WIDTH [1:0] | LD width | LD format | Valid bits |
|---|---|---|---|
| 2'b00 | 24 bits | PWM 8-bit + PAM 16-bit | PWM 8-bit + PAM 14-bit |
| 2'b01 | 24 bits | PAM 12-bit + PWM 12-bit | PAM 10-bit + PWM 11-bit |
| 2'b10 | 24 bits | PAM 12-bit + PWM 12-bit | PAM 10-bit + PWM 12-bit |
| 2'b11 | 16 bits | LD 16-bit | LD 16-bit |

Table 1. LD data size according to LD_WIDTH

The burst size unit means the amount of LD data corresponding to one line and the one line of LD data corresponds to the total LD data of all activated channels, and the number of active channel is determined by the CH_SIZE value of the ENABLE CHANNEL 2 register. Here, the CH_SIZE value should be equal to the number of activated channels by the CHx_EN of the ENABLE CHANNEL 1/2 registers.

Burst size unit = 1-line LD data = CH_SIZE * LD data size

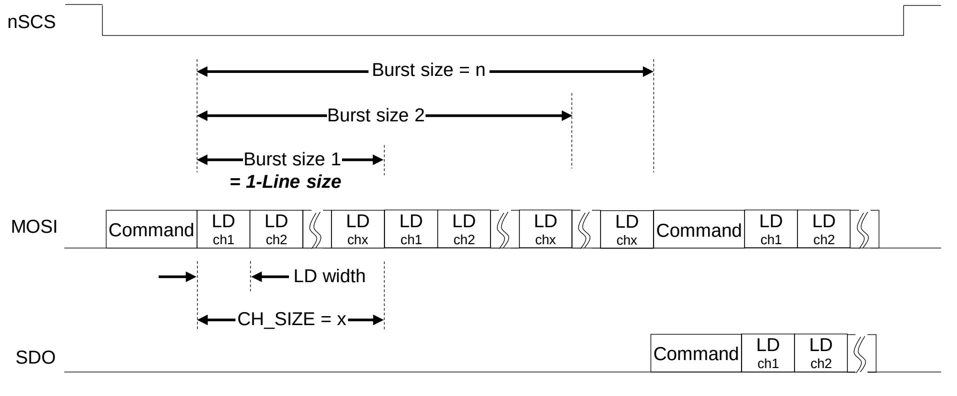



Figure 5. The SPI LD transfer in daisied 2 devices

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# Functional Description

## 2. Serial Interface

The XC24 has a total of 24 channels for serial daisy connections with a total of 31 XD12 devices. Each channel can transmit the data to the XD12 and receive the data from the XD12 through one line daisied to the XD12 devices. The XD24 has a total of 5 instructions for transferring the data to the XD12 and 3 instructions for receiving the data from the XD12.

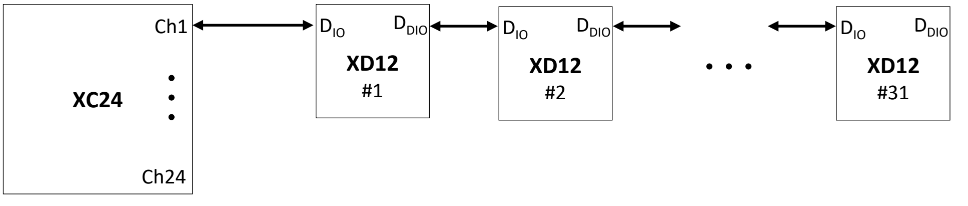


Figure 6. The daisied scheme

The commands are listed as shown in the Table 2. and described in more detail below.

| Instruction | Transaction direction | | Description |
|---|---|---|---|
| | Command | Data | |
| Global write | XC → XD | XC → XD | Global-write command the same register data between the XD devices |
| Local write | XC → XD | XC → XD | Local-write command the different register data between the XD devices |
| LD transfer | XC → XD | XC → XD | Local Dimming data-transfer command to the XD devices |
| ID gen | XC → XD | - | ID generation command of the XD devices in daisied each channel |
| Sync gen | XC → XD | - | Sync-generation command to the XD devices |
| Sync auto-gen | XC → XD | - | Automatically sync-generation command sync on the XD devices |
| | | | |
| Local read | XC → XD | XD → XC | Local-read command to the XD devices and read-back from the XD devices |
| Fault read | XC → XD | XD → XC | Fault-read command to the XD devices and read-back from the XD devices |
| Fault auto-read | XC → XD | XD → XC | Automatically fault-read command to the XD devices and read-back from the XD devices |

Table 2. Instruction list

# Functional Description

## ID-generation

The ID-GEN COMMAND register is used to set a unique ID for each device and each device's ID helps identifying when each device should react on a read-back operation. Therefore, after POR this command should be transacted at the first time for proper operation of the XD12 devices before transacting other commands. If the START bit of ID-GEN COMMAND register is set, then all activated channels begin issuing the command signals to all XD12 daisied in each channel and the command signal streams repeat continuously for the number of daisied devices.
The number of daisied devices is determined by the DAISY SIZE register of each channel. After the command signal streams are completed, all devices will have their own ID respectively. The ID is numbered in opposite direction of daisied line. It is careful that the writing data is transferred from the first located device, the last number ID, to the last located device, the first number ID, in daisied line but the reading data is received from the last located device to the first located device sequentially.

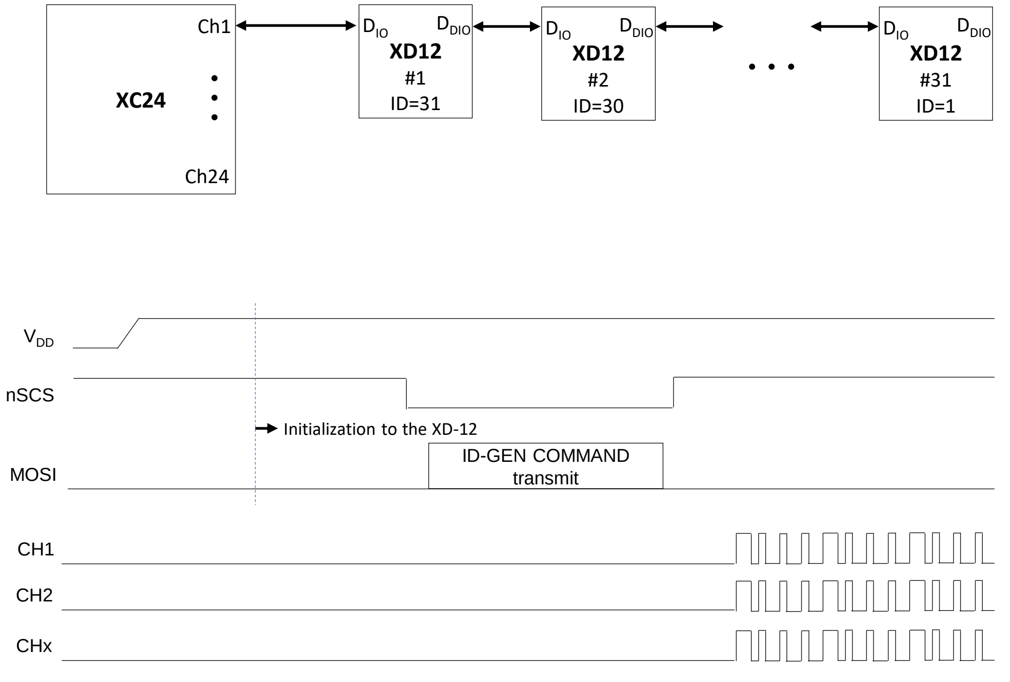


Figure 7. The sequence of ID-GEN command

# Functional Description

## Global-write

The Global-write command is used to write the register data to the XD12 only if all XD12 devices daisied to a channel of the XC24 have the same register data. If the register data to be written to the XD12 are different each other, at that time the Local-write command should be used. For executing the Global-write command properly, the GLOBAL-WRITE DATA register should be firstly written with desired value and then the START bit of the GLOBAL-WRITE COMMAND register should be set with desired address in order to begin transacting this command.

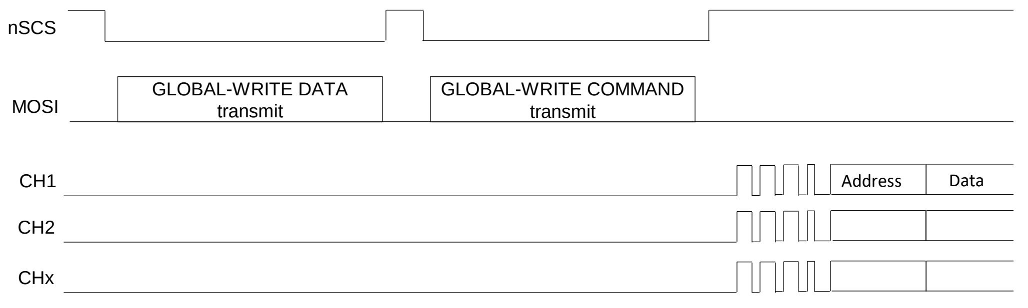


Figure 8. The sequence of Global-write command

The GLOBAL-WRITE COMMAND register consists of 6 bits of ADDRESS field and 1 bit of START. The ADDRESS is the register address of the XD12 to be accessed. If the START bit is set with the ADDRESS together, the Global writing command begins issuing simultaneously in all activated channels, and the command streams repeat continuously for the number of daisied XD12. The number of daisied devices can be determined by the DAISY SIZE register of each channel. The max number of daisied devices per channel is 31.

# Functional Description

## Local-write

The Local-write command is used to write the register data to the XD12 in case that all XD12 devices daisied to a channel of the XC24 have the different register data each other. If the register data to be written to the XD12 are same, at that time the Global-write command is more useful for fast transaction time. For proper execution of the Local-write command, the LOCAL-RW DATA registers should be firstly written with desired value and then, the START bit of the LOCAL-WRITE COMMAND register should be set with desired address for beginning this command. It is careful that the LOCAL-RW POINTER RESET register must be set necessarily to 1 for the next transaction after the current Local - write command execution is completed or before the next Local-write command will start again.

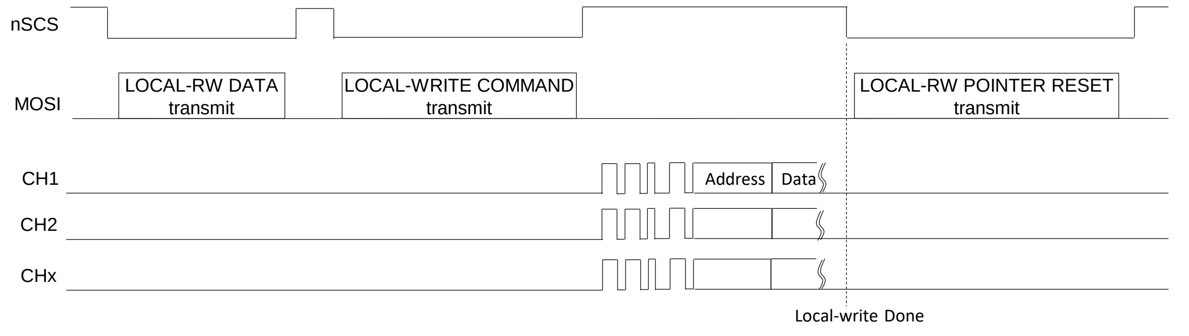


Figure 9. The sequence of Local-write command

The LOCAL-WRITE COMMAND register consists of ADDRESS 6 bits, CH_SEG 2 bits and START 1 bit. The ADDRESS is the register address of the XD12 to be accessed. If the START bit is set with the ADDRESS together, the Local-write command begins the execution simultaneously in all activated channels, and the command streams repeat continuously for the number of daisied XD12. The number of daisied devices can be determined by the DAISY SIZE register of each channel. The max number of daisied devices per channel is 31. The channel segment bits, CH_SEG, are used to decide on which channels are activated for this command, because this command can be performed 8 channels at a time.
The LOCAL-RW DATA registers are divided into a total of 8 port group and each port has a total of 16 registers. The first data register in the port is mapped to the first located XD12 in daisied order and the second data register is mapped to the second located XD12, and so on. Although the data registers in a port are a total 16, the XC-24 can support over 16 daisied devices in a channel by using the LOCAL-WR TRANSFER POINTER and LOCAL-RW DIFFERENCE POINTER. The LOCAL-WR TRANSFER DIFF POINTER of the LOCAL-RW DIFFERENCE POINTER shows how many data remain in the LOCAL-RW DATA buffers. A value of 0 means that the LOCAL-RW DATA buffers are empty, so you must continue to fill up before they are empty. A value greater than 16 means that the buffer has been overwritten, so you must wait until the pointer value is less than 16.

| CH_SEG[1:0] | Channel to be activated | LOCAL-RW DATA 1 ~ 16 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | PORT1 | PORT2 | PORT3 | PORT4 | PORT5 | PORT6 | PORT7 | PORT8 |
| 2’b00 | Ch 1 ~ Ch 8 | Ch 1 | Ch 2 | Ch 3 | Ch 4 | Ch 5 | Ch 6 | Ch 7 | Ch 8 |
| 2’b01 | Ch 9 ~ Ch 16 | Ch 9 | Ch 10 | Ch 11 | Ch 12 | Ch 13 | Ch 14 | Ch 15 | Ch 16 |
| 2’b10 | Ch 17 ~ Ch 24 | Ch 17 | Ch 18 | Ch 19 | Ch 20 | Ch 21 | Ch 22 | Ch 23 | Ch 24 |

Table 3. The LOCAL-RW DATA mapping according to the CH_SEG

# Functional Description

## Local-RW DATA structure for access

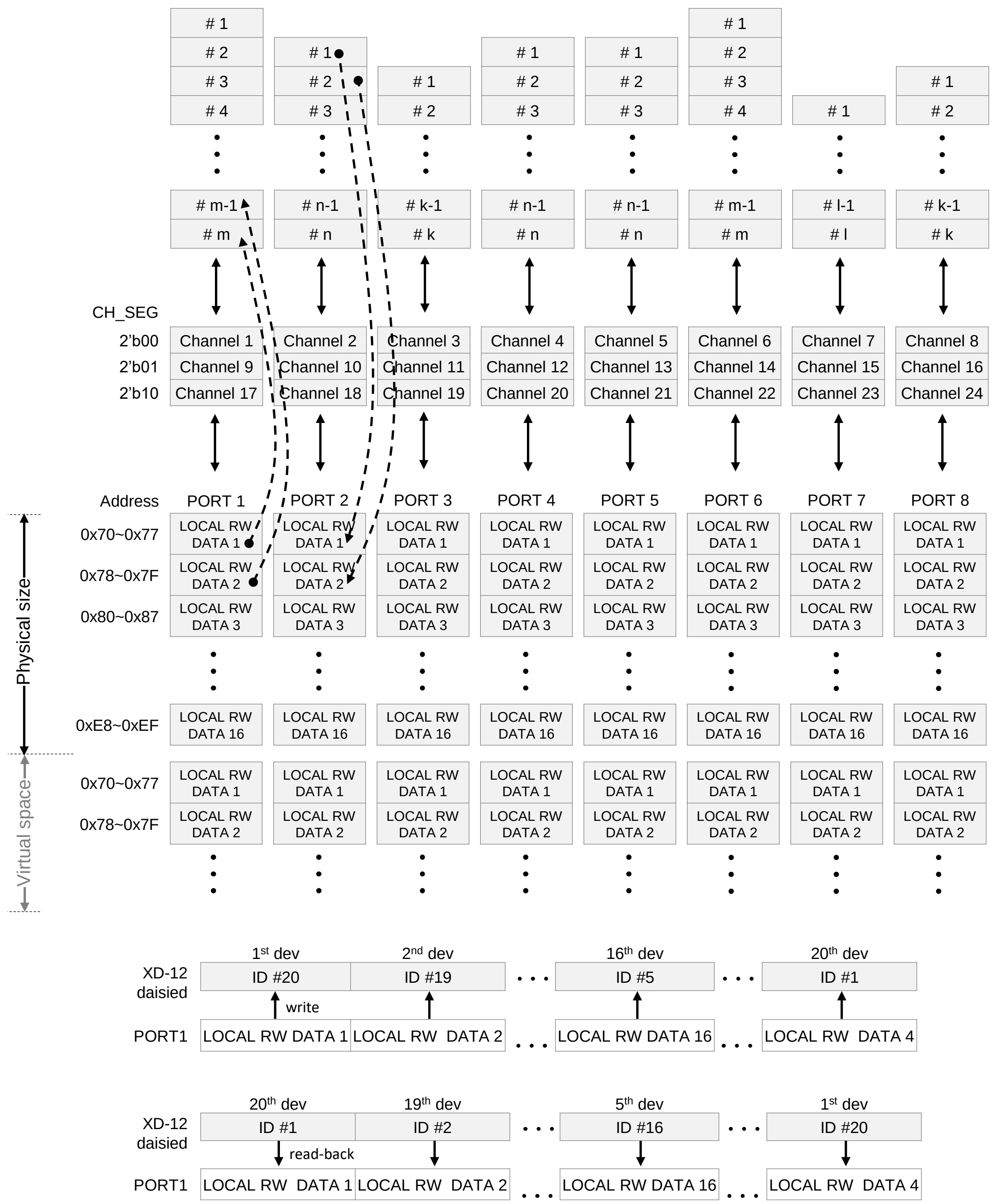


Figure 10. LOCAL-RW DATA mapping structure and data flow

# Functional Description

## Sync-generation

The Sync-generation command is used to synchronize the frame sync time for all daisied devices simultaneously. This command execution can be done by two methods, the one is by setting the START bit of the SYNC-GEN COMMAND register and the other is by setting the SYNC_AUTO_EN bit of the COMMAND AUTO ENABLE register.
If the SYNC_AUTO_EN bit is set, the Sync-generation command is automatically executed at the moment of receiving the VSYNC input. It is recommended to use this function, but it should be careful that The synchronization time of the XD12 devices may be as late as the number of XD12 devices than the time the VSYNC was received, so the VSYNC should arrive as soon as it is late. This command streams repeat continuously for the number of daisied devices in each channel.

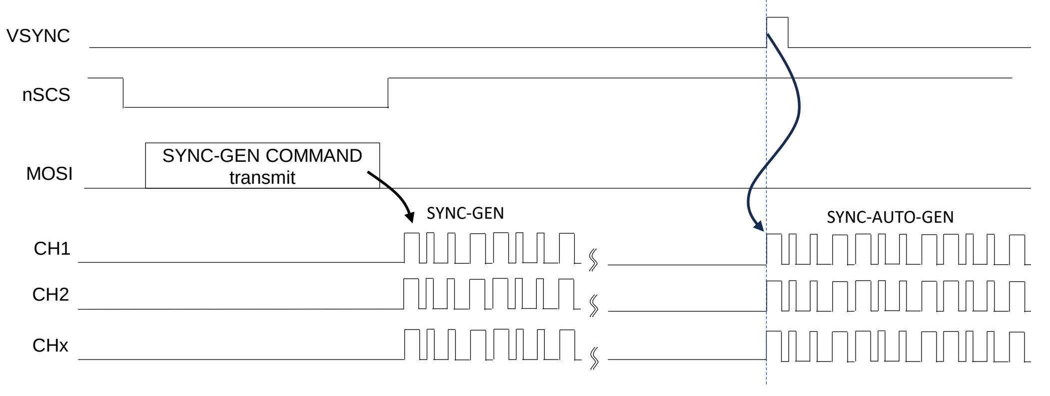


Figure 11. The sequence of SYNC-GEN command

## Fault-read

The Fault-read command is used to read-out the fault information of all daisied devices simultaneously. This command execution can be done by two methods, the one is by setting the START bit of the FAULT-GEN COMMAND register and the other is by setting the FAULT_AUTO_EN bit of the COMMAND AUTO ENABLE register.
If the FAULT_AUTO_EN bit is set, the Fault-read command is automatically executed after the completion of the Sync auto-gen command at the moment of receiving the VSYNC input. The fault data to be read-back from all XD-12 devices in each channel is stored into the corresponding channel's GLOBAL FAULT-READ DATA register, respectively.

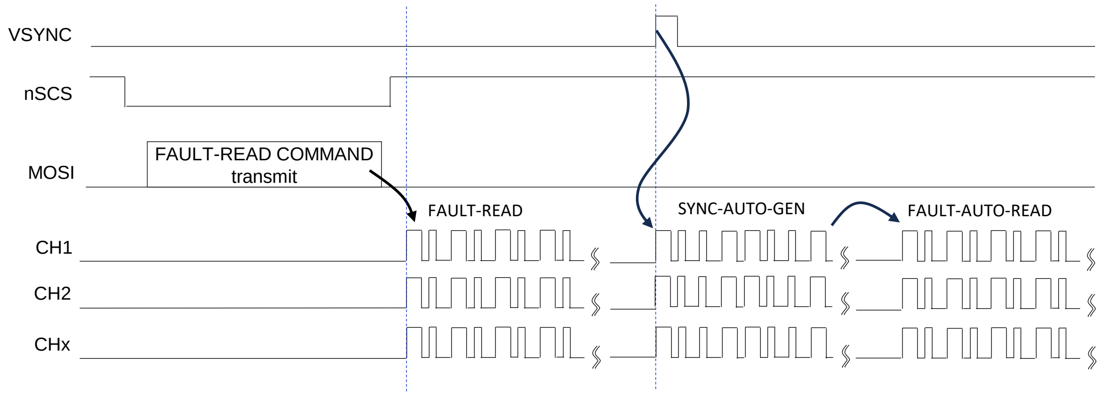


Figure 12. The sequence of FAULT-READ command

# Functional Description

## Local-read

The Local-read command is used to read the register data from all XD12 devices daisied in a channel of the XC24. Unlike the Local-write or Global-write command, the Local-read command can be used to read the register data of the XD12 devices regardless of whether the data is the same or different. The data to be read-out from the XD12 devices are stored into the LOCAL-RW DATA registers sequentially like the Figure 10. The START bit of the LOCAL-READ COMMAND register should be set with desired address to be read-out for beginning this command. It is careful that the LOCAL-RW POINTER RESET register must be set necessarily to 1 for the next transaction after the current Local read command execution is completed or before the next Local-read command will start again.

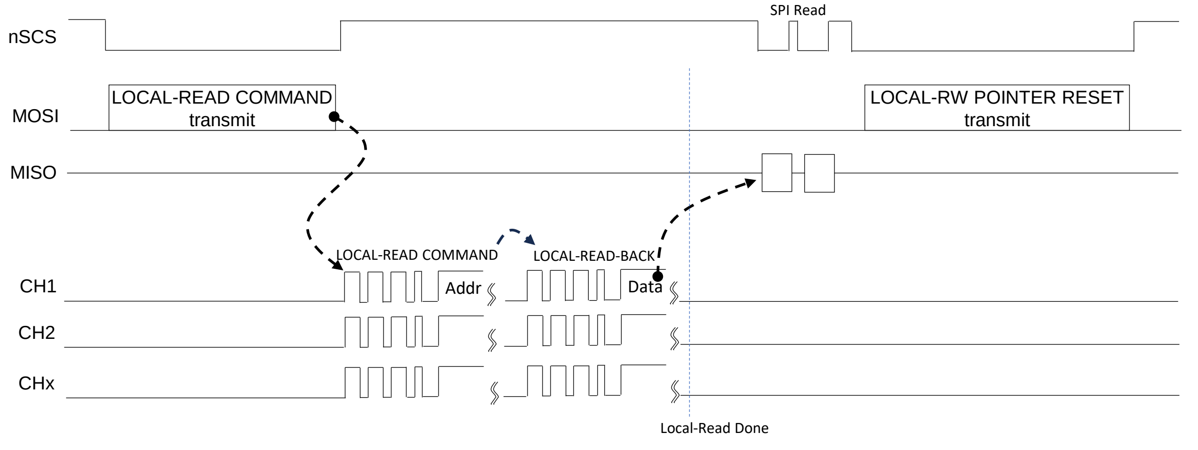


Figure 13. The sequence of Local-read command

The LOCAL-READ COMMAND register consists of ADDRESS 6 bits, CH_SEG 2 bits and START 1 bit. The ADDRESS is the register address of the XD12 to be accessed. If the START bit is set with the ADDRESS together, the Local-read command begins the execution simultaneously in all activated channels, and the command streams repeat continuously for the number of daisied XD12. The number of daisied devices can be determined by the DAISY SIZE register of each channel. The max number of daisied devices per channel is 31. The channel segment bits, CH_SEG, are used to decide on which channels are activated for this command, because this command can be performed 8 channels at a time.
The LOCAL-RW DATA registers are divided into a total of 8 port group and each port has a total of 16 registers. Unlike the Local-write command, the first data register in the port is mapped to the last located XD-12 in daisied order and the second data register is mapped to the second located XD-12 from the back, and so on. Although the data registers in a port are a total 16, the XC-24 can support over 16 daisied devices in a channel by using the LOCAL-READ RECEIVE POINTER and LOCAL-RW DIFFERENCE POINTER. The LOCAL-RD RECEIVE DIFF POINTER of the LOCAL-RW DIFFERENCE POINTER shows how many data are filled up in the LOCAL-RW DATA buffers. A value of 0 means that the LOCAL-RW DATA buffers are empty, so you must wait for the buffers to be refilled, and if the value is greater than 16, it means that the LOCAL-RW DATA buffers have been overwritten, so you must drag out the buffers before they are overflowed. Refer to the Table 3. and the Figure 10. for the mapping scheme.

# Functional Description

## Sequence of Local Dimming data transfer

The Local Dimming data of the next frame should be transmitted to the XC24 through the SPI and the XC24 should complete to transfer the LD data to the XD12 devices before assertion of the next frame sync. And then, the LD data will be started dimming-out by the XC12 devices at the next frame sync. The LD_WR_POINTER of the LD WR POINTER register will be increased one by one in line unit whenever the LD data are written to the XC24 LD data buffers through the SPI and see the Figure 5. for the amount of one line LD. The LD_RD_POINTER will be increased one by one whenever the XC24 transmits the amount of one LD data to the XD12 devices. The LD_DIFF_POINTER is a value of the LD_WR_POINTER minus the LD_RD_POINTER, so the value is useful to check how many LD data remains in the internal LD buffers.
If the LD_TANS_START_POINTER value becomes equal to the LD_DIFF_POINTER, then the XC24 starts sending the LD data to the XD12 devices automatically until the LD_DIFF_POINTER value becomes to 0.

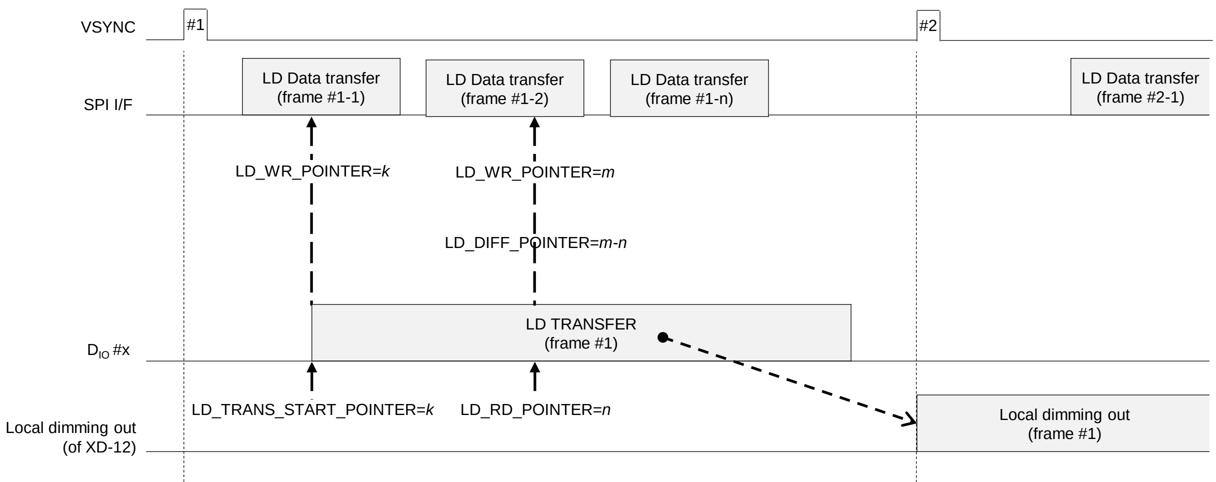


Figure 14. The sequence of Local dimming Data transfer

# Functional Description

## Local Dimming data buffer

The total number of Local Dimming data buffers is 32 per channel and 1 buffer can contain one block data which consists of PAM and PWM data. The size of PAM and PWM data depends on the LD_WIDTH of CHANNEL ENABLE 2 register and refer to previous chapter for more details. Even if the physical buffer size is limited to 32, the XC24 can support to transfer up to 511 blocks by using the LD WRITE POINTER, LD READ POINTER and LD DIFFERENCE POINTER. You can continue to send the LD data so that the LD DIFFERENCE POINTER does not become 0. The number of daisied devices of each channel can be different from each other by separately setting the DAISY SIZE register of each channel. The below shows the scheme of LD buffers and the mapping to the daisied device. For more information refer to the example on the next page.

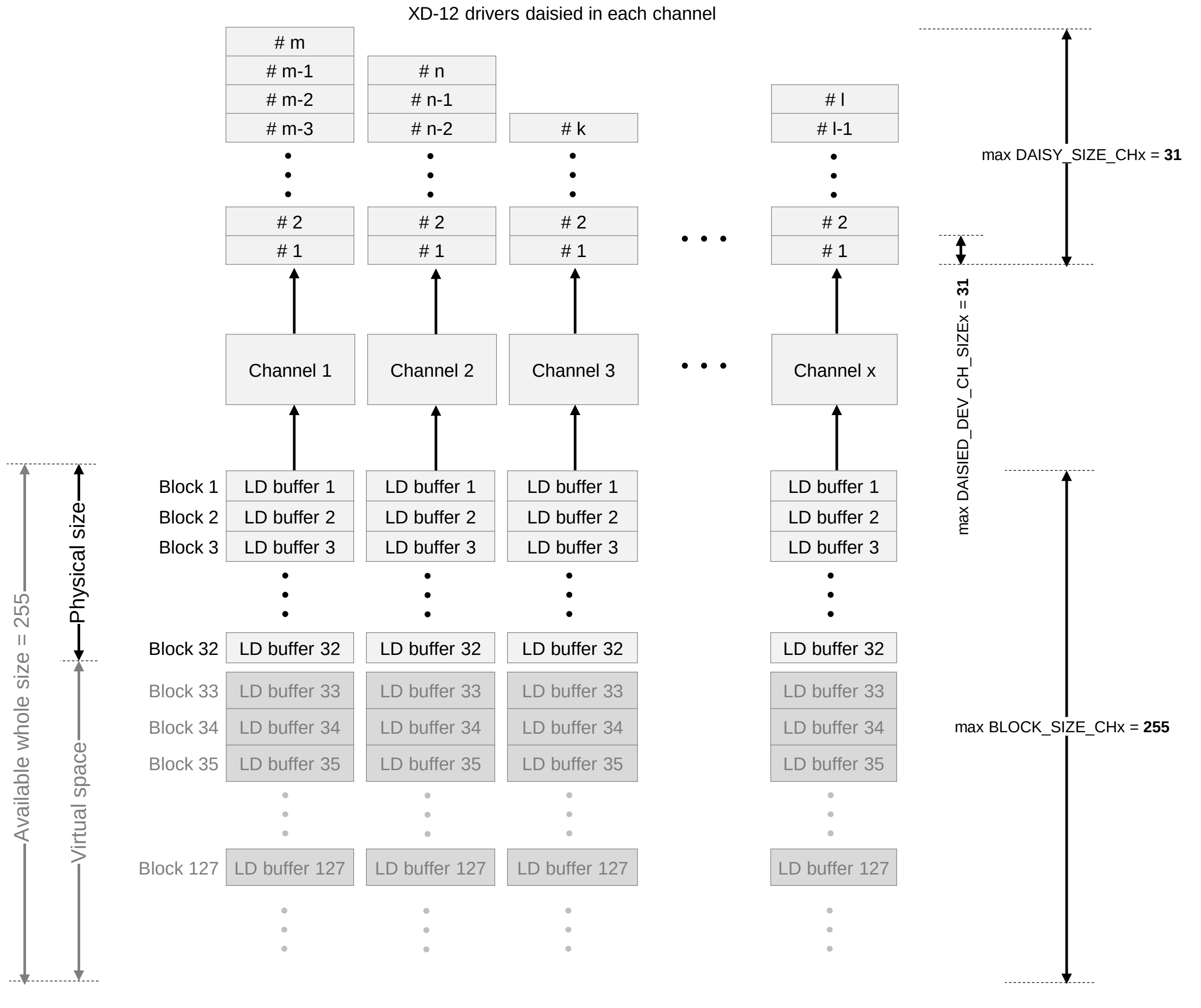


Figure 15. The scheme of Local dimming Data buffers

# Functional Description

## Local Dimming data transfer

The below example shows how to set properly the registers relevant to the Local Dimming data transfer. The CH_SIZE of the CHANNEL ENABLE 2 register determines how many channels among a total of 24 channels of XC24 will be activated, and if 20 channels will be used, then the CH_SIZE is 0x14 and available channels become from the channel 1 to channel 20, so it should be careful that the order of channels to be selected must be a direction that sequentially increases from 1 to 24. The DAISY_SIZE_CHx of the DAISY SIZE registers determines how many XD12 devices will be daisied in the channel x, and if 10 XD12 devices are daisied in a channel x, then the DAISY_ SIZE_CHx should be set to 0xA. The DAISIED_DEV_CH_SIZE determines how many channels of the XD12 will be used, so if the value is 0x4, the XD12 devices daisied in a channel x will capture only 4 LD data and then bypass the flowing LD data streams to the next XD12 devices and the remainder of the multiples of 4 be delivered lastly to a las XD12 device. It should be careful that all devices except for the last device must have the same value of DAISIED_ DEV_CH_SIZE. The BLOCK_SIZE_CHx determines the total number of LD data to be transferred to a channel x.

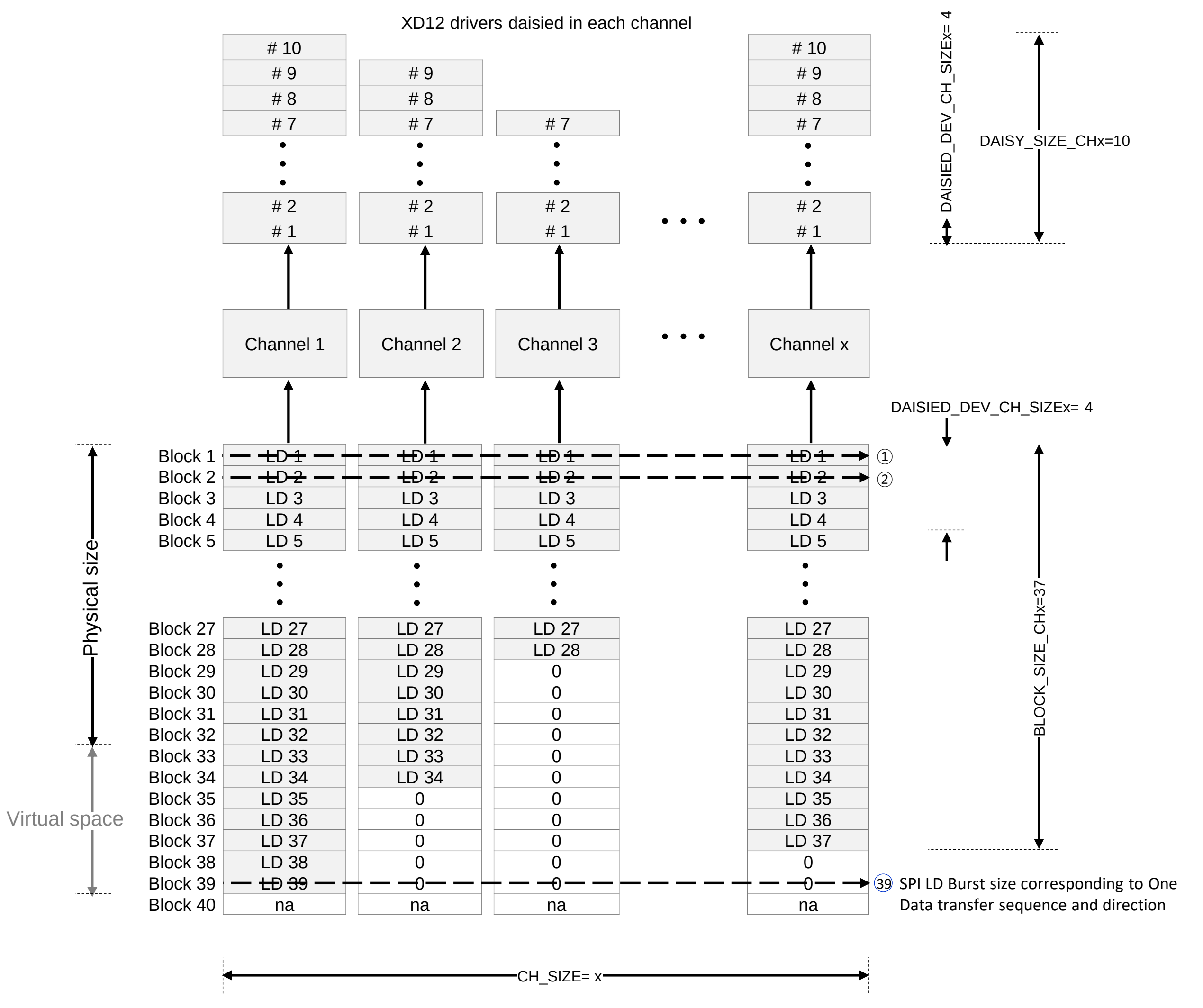


Figure 16. The example of Local dimming Data-transfer

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# 3. Register Map

The SPI interface protocol consists of two segments, the command 16-bit followed by the register's data 16-bit or a series of local dimming data.

## Command packet format

| B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| COMMAND CODE | | START ADDRESS [7:0] | | | | | | | | BURST SIZE [5:0] | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 15:14 | COMMAND CODE | 2'b00 : No operation.<br>2'b01 : Register Read.<br>2'b10 : Register Write.<br>2'b11 : Local dimming data transfer. |
| 13:6 | START ADDRESS | The start address of registers to access. |
| 5:0 | BURST SIZE | Amount of data to transfer or receive.<br>In case of accessing the registers, the number 1 means one packet of 16-bit data. In case of transferring the Local dimming data, the number 1 means one packet of (LD size * enabled channel numbers). The LD size and enabled channel numbers depend on the LD_WIDTH[1:0] and its bits of CHANNEL ENABLE 1 and 2 registers. |

## Data packet format

| B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DATA [15:0] | | | | | | | | | | | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 15:0 | DATA [15:0] | In case of COMMAND CODE :<br>2'b01 A register's data to read.<br>2'b10 A register's data to write.<br>2'b11 A series of local dimming data. Be careful that the data size and width to transfer the local dimming data are different from the register access. Refer to previous chapter 2. SPI protocol |

## SOFT RESET

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h00 | 0x0 | x | x | x | x | x | x | x | x | x | x | VS_RST2 | VS_RST1 | x | RST3 | RST2 | RST1 |

| Bits | Fields | Descriptions |
|---|---|---|
| 5 | VS_RST2 | If this bit is set, all de-serializers’ sm are reset at VSYNC rising. Normally, no need to use this bit. |
| 4 | VS_RST1 | If this bit is set, all serializers’ sm are reset at VSYNC rising. Normally, no need to use this bit. |
| 2 | RST3 | Reset to all types of buffer, the frame buffers, the global and local r/w registers and the global fault read registers. Auto-cleared. Normally, no need to use this bit. |
| 1 | RST2 | Reset to all serializers’ and de-serializers’ state machine, Auto-cleared. Normally, no need to use this bit. |
| 0 | RST1 | Reset to all logics, Auto-cleared (No need to clear after set) It need to set this bit after POR to ensure XC-24 reset. |

## GLOBAL-WRITE COMMAND

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h01 | 0x0 | START | x | x | x | x | x | x | x | x | x | ADDR [5:0] | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 15 | START | ‘0’ No operation.<br>‘1’ Start executing the global-write command with the GLOBAL WRITE DATA register, together, and this bit is auto-cleared after completing this execution. For correct operation, It needs to set the GLOBAL WRITE DATA register with desired value before setting this bit. The global-writing can be used in case that the target registers’ values of all XD-12 daisied devices are all the same. If the values are different from each other, then the local-write command should be used instead of the global-write command. |
| 5:0 | ADDRESS [5:0] | The same target address of all XD-12 daisied devices to write. |

## LOCAL-WRITE COMMAND

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h02 | 0x0 | START | x | x | x | x | x | CH_SEG [1:0] | | x | x | ADDR [5:0] | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 15 | START | ‘0’ No operation.<br>‘1’ Start executing the local-write command with the LOCAL RW DATA registers together, and this bit is auto-cleared after completing this execution. For correct operation, It needs to set the LOCAL RW DATA registers with desired values before setting this bit. The local-write command is used in case that the target registers’ values of all XD-12 daisied devices are different from each other. |
| 9:8 | CH_SEG [1:0] | These bits are used to designate which channels are activated for this command.<br>2’b00 The channel 1 from 8 are activated.<br>2’b01 The channel 9 from 16 are activated.<br>2’b10 The channel 17 from 24 are activated.<br>2’b11 No allocated.<br>Refer to previous chapter. how to allocate the LOCAL RW DATA registers according to CH_SEG [1:0]. |
| 5:0 | ADDRESS [5:0] | The same target address of all XD-12 daisied devices to write |

## LOCAL-READ COMMAND

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h03 | 0x0 | START | x | x | x | x | x | CH_SEG [1:0] | | x | x | ADDR [5:0] | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 15 | START | ‘0’ No operation.<br>‘1’ Start executing the local-read command with the LOCAL RW DATA registers together, and this bit is auto-cleared after completing this execution. For correct operation, It needs to set the LOCAL RW DATA registers with desired values before setting this bit. The local-read command is used to read out the target registers’ values of all XD-12 daisied devices. |
| 9:8 | CH_SEG [1:0] | These bits are used to designate which channels are activated for this command<br>2’b00 : The channel 1 from 8 are activated.<br>2’b01 : The channel 9 from 16 are activated.<br>2’b10 : The channel 17 from 24 are activated.<br>2’b11 : No allocated.<br>Refer to previous chapter. how to allocate the LOCAL RW DATA registers according to CH_SEG [1:0]. |
| 5:0 | ADDRESS [5:0] | The same target address of all XD-12 daisied devices to read out. |

## ID-GEN COMMAND

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h04 | 0x0 | START | x | x | x | x | x | x | x | x | x | x | x | x | x | x | x |

| Bits | Fields | Descriptions |
|---|---|---|
| 15 | START | ‘0’ No operation.<br>‘1’ Start executing the id-generating command to all XD-12 daisied devices and this bit is auto-cleared after completing this execution. All XD-12 daisied devices connected in each channel will have their own ID number after completing the id-generating command. This command is needed to read-out the XD-12 drivers’ registers data correctly in sequential, so, this command should be executed before executing the local-reading or the fault-reading command. It is recommended that the id-generating command should be executed firstly on system initialization. |

## FAULT-READ COMMAND

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h05 | 0x0 | START | x | x | x | x | x | x | x | x | x | x | x | x | x | x | x |

| Bits | Fields | Descriptions |
|---|---|---|
| 15 | START | ‘0’ No operation.<br>‘1’ Start executing the fault-reading command to all XD-12 daisied devices to monitor the fault event like as the thermal occurrence, the short status, the open status and the fb status of the XD-12 devices and this bit is auto-cleared after completing this execution. This command can be also auto-executed at the VSYNC rising edge if the FAULT_ AUTO_EN bit in the COMMAND AUTO ENABLE register is set. |

## LD-TRANSFER COMMAND

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h06 | 0x0 | START | x | x | x | x | x | x | x | x | x | x | x | x | x | x | x |

| Bits | Fields | Descriptions |
|---|---|---|
| 15 | START | ‘0’ No operation.<br>‘1’ Start executing the local dimming data transfer command to all XD-12 daisied devices in order to transfer the LD data in burst from the XC-24 to the XD-12 drivers and this bit is auto-cleared after the completion. This command can be also auto-executed immediately after the amount of LD data to be transferred to the XC-24 through the SPI interface is reached to the value of the LD_RD_START_POINTER of the LD START POINTER/TH register. |

## SYNC-GEN COMMAND

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h07 | 0x0 | START | x | x | x | x | x | x | x | x | x | x | x | x | x | x | x |

| Bits | Fields | Descriptions |
|---|---|---|
| 15 | START | '0' No operation.<br>'1' Start executing the sync-gen command to all XD12 daisied devices in order to make the frame synchronization to the drivers and this bit is auto-cleared after this execution. This command can be also auto-executed at the VSYNC rising edge, if the SYNC_ AUTO_EN bit in the COMMAND AUTO ENABLE register is set. It is recommended that the SYNC_AUTO_EN bit is set for good local dimming. |

## COMMAND AUTO ENABLE

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h08 | 0x0 | x | x | x | x | x | x | x | TIMEOUT_EN | x | x | x | FAULT_AUTO_EN | x | x | x | SYNC_AUTO_EN |

| Bits | Fields | Descriptions |
|---|---|---|
| 8 | TIMEOUT_EN | '0' No operation<br>'1' The timeout function is activated during the execution of read-back commands. The timer starts from the end of completing whole command frame data out corresponding to all daisied device to first bit arrived time. The timer is expired at (MCLK * 8,192). |
| 4 | FAULT_AUTO_EN | '0' N operation.<br>'1' Start executing the fault-reading command automatically. If this bit is set, the fault-reading command is auto-executed in two conditions. The first is occurred at the time of VSYNC rising edge and the second condition is occurred at the time of expiration of the FAULT AUTO READ TIMER register for the number of occurrence of the FAULT AUTO READ EVENT register. |
| 0 | SYNC_AUTO_EN | '0' No operation.<br>'1' Start executing the sync-gen command automatically. If this bit is set, the sync-gen command is auto-executed at the time of VSYNC rising edge. |

## LD-WRITE POINTER

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h0A | RO | x | x | x | x | x | x | x | LD_WR_POINTER [8:0] | | | | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 8:0 | LD_WR_POINTER | This register value represents the number of times the local dimming data are written into the internal LD buffers through the SPI interface. The number increases one by one whenever the local dimming data corresponding to (the number of enabled channels * one LD data width) are transferred to the internal LD buffers. This pointer value is reset at the VSYNC rising.<br><br>LD_WIDTH of the CHANNEL ENABLE 2 register :<br>2'b00 : one block LD data width = 24 bits<br>2'b01 : one block LD data width = 24 bits<br>2'b10 : one block LD data width = 24 bits<br>2'b11 : one block LD data width = 16 bits |

## LD-READ POINTER

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h0B | RO | x | x | x | x | x | x | x | LD_RD_POINTER [8:0] | | | | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 8:0 | LD_RD_POINTER | This register value represents the number of times the local dimming data are read-out to transfer the data from the internal LD buffers to the XD12 drivers. The number increases one by one whenever the local dimming data corresponding to the one block LD data of the XD12 drivers are read-out. This pointer value is reset at the VSYNC rising. |

## LD-DIFFERENCE POINTER

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h0C | RO | x | x | x | x | x | x | x | x | x | x | LD_DIFF_POINTER [5:0] | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 5:0 | LD_DIFF_POINTER | This value is useful to monitor how much the LD data remains in the internal LD buffers for checking overrun or underrun.<br>= LD_WR_POINTER – LD_RD_POINTER |

## LD-TRANSFER START POINTER / THRESHOLD

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h0D | 0x0 | x | x | LD_TRANS_START_POINTER [5:0] | | | | | | INT_LD_SIGN | x | LD_DIFF_THRESHOLD [5:0] | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 13:8 | LD_TRANS_START_POINTER | The LD_TRANS_START_POINTER determines when the transmit from the internal LD buffers to the XD12 drivers will begin. If the value is set to 8, then the transmit will begin as soon as the number of times the data through the SPI interface becomes 8. Once the transmit is stated, the transaction will be continued until the number of transmit reaches to the BLOCK_SIZE value of the BLOCK SIZE registers. |
| 7 | INT_LD_SIGN | Don't set this bit. |
| 5:0 | LD_DIFF_THRESHOLD | The LD_DIFF_THRESHOLD is used to set in which condition the nINT_LD output signal will be issued to the host. Whenever the LD_DIFF_POINTER is equal to or smaller than the LD_DIFF_THRESHOLD, then the nINT_LD signal will be issued. This feature helps to check how much LD data remains in the internal LD buffer for monitoring overruns or underruns of the LD data. |

## LOCAL-WR TRANSFER POINTER

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h0E | RO | x | x | LOCAL_WR_OUT_POINTER [5:0] | | | | | | x | x | LOCAL_WR_TRANS_POINTER [5:0] | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 12:8 | LOCAL_WR_OUT_POINTER | This register value represents the number of times the data from the LOCAL RW DATA registers are read-out to write the data to the XD12 drivers' registers. The number increases one by one whenever the data are read-out from the LOCAL RW DATA registers. |
| 5:0 | LOCAL_WR_TRANS_POINTER | This register value represents the number of times the data to be transferred to the XD12 drivers are written to the LOCAL RW DATA registers through the SPI interface. The number increased one by one whenever the data are written to the LOCAL RW DATA registers. |

## LOCAL-RD RECEIVE POINTER

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h0F | RO | x | x | LOCAL_RD_OUT_POINTER [5:0] | | | | | | x | x | LOCAL_RD_REC_POINTER [5:0] | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 12:8 | LOCAL_RD_OUT_POINTER | This register value represents the number of times the data transferred from the XD12 drivers are read out from the LOCAL RW DATA registers for transferring the data through the SPI interface. The number increases one by one whenever the data are read-out from the LOCAL RW DATA registers. |
| 5:0 | LOCAL_RD_REC_POINTER | This register value represents the number of times the data to be transferred through the SPI interface are written from the XD-12 drivers to the LOCAL RW DATA registers. The number increased one by one whenever the data are written to the LOCAL RW DATA registers. |

## LOCAL-RW DIFFERENCE POINTER

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h10 | RO | x | x | LOCAL_RD_DIFF_POINTER [5:0] | | | | | | x | x | LOCAL_WR_DIFF_POINTER [5:0] | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 12:8 | LOCAL_RD_DIFF_POINTER | This value is useful to monitor how much the data from the XD12 drivers remains in the LOCAL RW DATA registers for checking overrun or underrun.<br>= LOCAL_RD_REC_POINTER – LOCAL_RD_OUT_POINTER |
| 5:0 | LOCAL_WR_DIFF_POINTER | This value is useful to monitor how much the data to be transferred to the XD12 remains in the LOCAL RW DATA registers for checking overrun or underrun.<br>= LOCAL_WR_TRANS_POINTER – LOCAL_WR_OUT_POINTER |

## LOCAL-RW POINTER RESET

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h11 | 0x0 | x | x | x | x | x | x | x | LOCAL_RD_POINTER_RST | x | x | x | x | x | x | x | LOCAL_WR_POINTER_RST |

| Bits | Fields | | Descriptions |
|---|---|---|---|
| 8 | LOCAL_RD_POINTER_RST | '0'<br>'1' | No operation<br>The LOCAL_RD_OUT_POINTER and the LOCAL_RD_REC_POINTER are reset. This bit is auto-cleared. After completing the Local read command, the relevant pointer value must be cleared by this bit before the next Local read command. |
| 0 | LOCAL_WR_POINTER_RST | '0'<br>'1' | No operation<br>The LOCAL_WR_OUT_POINTER and the LOCAL_WR_TRANS_POINTER are reset. This bit is auto-cleared. After completing the Local write command, the relevant pointer value must be cleared by this bit before the next Local write command. |

## FAULT-AUTO-READ TIMER

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h12 | 0x0 | FAULT_AUTO_RD_TIMER [15:0] | | | | | | | | | | | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 15:0 | FAULT_AUTO_RD_INTERVAL | This register determines the execution interval for the fault auto-read commands. Counted in MCLK units. |

## FAULT-AUTO-READ EVENT

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h13 | 0x0 | FAULT_AUTO_RD_EVENT [7:0] | | | | | | | | x | x | x | x | x | x | x | FAULT_AUTO_RD_INTERVAL [16] |

| Bits | Fields | Descriptions |
|---|---|---|
| 15:8 | FAULT_AUTO_RD_EVENT | This register determines the execution times for the fault auto-read commands. The value is reset at VSYNC rising edge. |
| 0 | FAULT_AUTO_RD_INTERVAL | The last bit of the FAULT_AUTO_RD_TIMER [16:0] |

## SERIALIZER CLOCK GEN

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h14 | 0x0408 | x | x | x | x | SCK_LOW [3:0] | | | | x | x | x | SCK_HIGH [4:0] | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 11:8 | SCK_LOW | This register determines the high pulse duration in case the serial data-out drives the low state. Counted in MCLK units. |
| 4:0 | SCK_HIGH | This register determines the high pulse duration in case the serial data-out drives the high state. Counted in MCLK units. |

## INTERRUPT ENABLE

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h15 | 0x0 | x | x | x | x | x | x | x | INT_TIMEOUT_ERR_EN | INT_FAULT_RD_FAIL_EN | INT_FAULT_AUTO_RD_FAIL_EN |

| Addr | Default | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|
| 'h15 | 0x0 | INT_RD_REC_FAIL_EN | INT_LD_EN | INT_THERMAL_EN | INT_SHORT_EN | INT_OPEN_EN | INT_FB_EN |

| Bits | Fields | Descriptions | |
|---|---|---|---|
| 8 | INT_TIMEOUT_ERR_EN | '1' | This bit enables to issue the interrupt through the nINT_FAULT pin in case that the timer will be expired during the execution for read-back commands. The expired time is mclk * 8,192. |
| 7 | INT_FAULT_RD_FAIL_EN | '1' | This bit enables to issue the interrupt through the nINT_FAULT pin in case that any violation happens during the execution for the fault read-back from the XD12. |
| 6 | INT_FAULT_AUTO_RD_FAIL_EN | '1' | This bit enables to issue the interrupt through the nINT_FAULT pin in case that any violation happens during the execution for the fault auto-read-back from the XD12. |
| 5 | INT_RD_REC_FAIL_EN | '1' | This bit enables to issue the interrupt through the nINT_FAULT pin in case that any violation happens during the execution for the local read-back from the XD12. |
| 4 | INT_LD_EN | '1' | This bit enables to issue the interrupt through the nINT_FAULT pin in case that the LD_DIFF_POINTER is equal to or smaller than the LD_DIFF_THRESOLD during the execution of the LD transfer to the XD12. |
| 3 | INT_THERMAL_EN | '1' | This bit enables to issue the interrupt through the nINT_FAULT pin in case that the thermal regulation happens among the XD12 drivers during dimming operation. |
| 2 | INT_SHORT_EN | '1' | This bit enables to issue the interrupt through the nINT_FAULT pin in case that the short condition happens among the XD12 drivers during dimming operation. |
| 1 | INT_OPEN_EN | '1' | This bit enables to issue the interrupt through the nINT_FAULT pin in case that the open condition happens among the XD12 drivers during dimming operation. |
| 0 | INT_FB_EN | '1' | This bit enables to issue the interrupt through the nINT_FAULT pin in case that the fb condition happens among the XD12 drivers during dimming operation. |

## COMMAND STATUS 1

| Addr | Default | B15 | B14 | B13 | B12 |
|---|---|---|---|---|---|
| ‘h16 | RO | LOCAL_RD_DONE | LOCAL_RD_DOING | LOCAL_WR_DONE | LOCAL_WR_DOING |

| Addr | Default | B11 | B10 | B9 | B8 | B7 | B6 |
|---|---|---|---|---|---|---|---|
| ‘h16 | RO | GLOBAL_WR_DONE | GOLBAL_WR_DOING | LD_TRANS_DONE | LD_TRANS_DOING | FAULT_DONE | FAULT_DOING |

| Addr | Default | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|
| ‘h16 | RO | FAULT_AUTO_DONE | FAULT_AUTO_DOING | SYNC_AUTO_DONE | SYNC_AUTO_DOING | SYNC_DONE | SYNC_DOING |

| Bits | Fields | Descriptions |
|---|---|---|
| 15 | LOCAL_RD_DONE | This bit is set after the execution for the local-reading command is completed. |
| 14 | LOCAL_RD_DOING | This bit is set while the local-reading command is working until it is complete. |
| 13 | LOCAL_WR_DONE | This bit is set after the execution for the local-writing command is completed. |
| 12 | LOCAL_WR_DOING | This bit is set while the local-writing command is working until it is complete. |
| 11 | GLOBAL_WR_DONE | This bit is set after the execution for the global-writing command is completed. |
| 10 | GLOBAL_WR_DOING | This bit is set while the global-writing command is working until it is complete. |
| 9 | LD_TRANS_DONE | This bit is set after the execution for the LD transferring command is completed. |
| 8 | LD_TRANS_DOING | This bit is set while the LD transferring command is working until it is complete. |
| 7 | FAULT_DONE | This bit is set after the execution for the fault-reading command is completed. |
| 6 | FAULT_DOING | This bit is set while the fault-reading command is working until it is complete. |
| 5 | FAULT_AUTO_DONE | This bit is set after the execution for the fault auto-reading command is completed. |
| 4 | FAULT_AUTO_DOING | This bit is set while the fault auto-reading command is working until it is complete. |
| 3 | SYNC_AUTO_DONE | This bit is set after the execution for the sync auto-gen command is completed. |
| 2 | SYNC_AUTO_DOING | This bit is set while the sync auto-gen command is working until it is complete. |
| 1 | SYNC_DONE | This bit is set after the execution for the sync-gen command is completed. |
| 0 | SYNC_DOING | This bit is set while the sync-gen command is working until it is complete. |

## COMMAND STATUS 2

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h17 | RO | x | x | x | x | x | x | x | x | x | x | x | x | x | x | ID_GEN_DONE | ID_GEN_DOING |

| Bits | Fields | Descriptions |
|---|---|---|
| 1 | ID_GEN_DONE | This bit is set after the execution for the id-gen command is completed. |
| 0 | ID_GEN_DOING | This bit is set while the id-gen command is working until it is complete. |

## RECEIVE STATUS

| Addr | Default | B15 | B14 | B13 | B12 |
|---|---|---|---|---|---|
| ‘h18 | RO | x | FAULT_REC_FAIL | FAULT_AUTO_REC_FAIL | LOCAL_REC_FAIL |

| Addr | Default | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 |
|---|---|---|---|---|---|---|---|---|---|
| ‘h18 | RO | x | x | x | TIMEOUT_ERR | x | x | FAULT_REC_DONE | FAULT_REC_DOING |

| Addr | Default | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|
| ‘h18 | RO | FAULT_AUTO_REC_DONE | FAULT_AUTO_REC_DOING | LOCAL_REC_DONE | LOCAL_REC_DOING |

| Bits | Fields | Descriptions |
|---|---|---|
| 14 | FAULT_REC_FAIL | This bit is set if any violation happens while the fault-receiving job is working. |
| 13 | FAULT_AUTO_REC_FAIL | This bit is set if any violation happens while the fault auto-receiving job is working. |
| 12 | LOCAL_REC_FAIL | This bit is set if any violation happens while the local read-receiving job is working. |
| 8 | TIMEOUT_ERR | This bit is set if the timer will be expired while the execution of read-back due to no response from the XD devices. |
| 5 | FAULT_REC_DONE | This bit is set after the fault-receiving job is completed. |
| 4 | FAULT_REC_DOING | This bit is set while the fault-receiving job is working until it is complete. |
| 3 | FAULT_AUTO_REC_DONE | This bit is set after the fault auto-receiving job is completed. |
| 2 | FAULT_AUTO_REC_DOING | This bit is set while the fault auto-receiving job is working until it is complete. |
| 1 | LOCAL_REC_DONE | This bit is set after the local read-receiving job is completed. |
| 0 | LOCAL_REC_DOING | This bit is set while the local read-receiving job is working until it is complete. |

## INTERRUPT STATUS

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 |
|---|---|---|---|---|---|---|---|---|
| ‘h19 | RO | x | x | x | INT_TIMEOUT_ERR_SRC | INT_FAULT_REC_FAIL_SRC | INT_FAULT_AUTO_REC_FAIL_SRC | INT_LOCAL_REC_FAIL_SRC |

| Addr | Default | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|
| ‘h19 | RO | INT_LD_TRAN_SRC | INT_TML_SRC | INT_SHORT_SRC | INT_OPEN_SRC | INT_FB_SRC | x | x | INT_LD | INT_FAULT |

| Bits | Fields | Descriptions |
|---|---|---|
| 12 | INT_TIMEOUT_ERR_SRC | This bit shows whether the timeout has been happened or not while the read-back commands are working until it is complete. This bit value will remain at the same level even if this register is read-out until the next command will be start. |
| 11 | INT_FAULT_REC_FAIL_SRC | This bit shows whether any violation has been happened or not while the fault-receiving job is working until it is complete. This bit value will remain at the same level even if this register is read-out until the next command will be start. |
| 10 | INT_FAULT_AUTO_REC_FAIL_SRC | This bit shows whether any violation has been happened or not while the fault auto-receiving job is working until it is complete. This bit value will remain at the same level even if this register is read-out until the next command will be start. |
| 9 | INT_LOCAL_REC_FAIL_SRC | This bit shows whether any violation has been happened or not while the local read-receiving job is working until it is complete. This bit value will remain at the same level even if this register is read-out until the next command will be start. |
| 8 | INT_LD_TRANS_SRC | This bit shows whether the LD interrupt has been asserted or not while transferring the LD data to XD12 drivers through the XC-24 controller. This bit value will remain at the same level even if this register is read-out until the next command will be start. |
| 7 | INT_TML_SRC | This bit shows the thermal status from all XD12 daisied to enabled channels after completing the fault (auto-) reading command. This bit value will remain at the same level even if this register is read-out until the next command will be start. |
| 6 | INT_SHORT_SRC | This bit shows the short status from all XD12 daisied to enabled channels after completing the fault (auto-) reading command. This bit value will remain at the same level even if this register is read-out until the next command will be start. |
| 5 | INT_OPEN_SRC | This bit shows the open status from all XD12 daisied to enabled channels after completing the fault (auto-) reading command. This bit value will remain at the same level even if this register is read-out until the next command will be start. |
| 4 | INT_FB_SRC | This bit shows the fb status from all XD12 daisied to enabled channels after completing the fault (auto-) reading command. This bit value will remain at the same level even if this register is read-out until the next command will be start. |
| 1 | INT_LD | This high-active bit indicates the nINT_LD pin has been asserted to the low-active level. The INT_LD represents an interrupt sources, INT_LD_TRAN_SRC, if its interrupt enable bits is set. If the INT_LD_EN bit is set and INT_LD_TRAN_SRC is asserted, then, this bit will be asserted through nINT_LD pin. This bit is auto-cleared after this INTERRUPT STATUS register is read-out. |
| 0 | INT_FAULT | This high-active bit indicates the nINT_FAULT pin has been asserted to the low-active level. The INT_FAULT represents several interrupt sources, INT_FB_SRC, INT_OPEN_SRC, INT_SHORT_SRC, INT_TML_SRC, INT_LOCAL_REC_FAIL_SRC, INT_FAULT_AUTO_REC_FAIL_SRC and INT_FAULT_REC_FAIL_SRC, if their own interrupt enable bits are set. If the corresponding interrupt enable bit is set and the interrupt source is asserted, then, this bit will be asserted through nINT_FAULT pin. This bit is auto-cleared after this INTERRUPT STATUS register is read-out. |

## FB PWM PERIOD

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h1A | 0x0 | FB_PWM_EN | FB_PWM_POL | T_FB_EN | x | x | FB_PWM_PERIOD [10:0] | | | | | | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 15 | FB_PWM_EN | If this bit is set, the PWM signal determined by the FB_PWM_PERIOD will be driven on the external FB pin. If this bit is '0', the signal, low or high, will be out to the external FB pin at the period of VSYNC or the FAULT_AUTO_READ_TIMER. |
| 14 | FB_PWM_POL | This bit determines a polarity of the FB signal toward the external FB pin. If this bit '0', the FB assertion will be showed in a low level on the FB pin because the pin has an open-drain property. |
| 13 | T_FB_EN | If this bit '1', the FB signal toward the FB pin will be de-asserted as long as the thermal event is active, even though the FB event happens. |
| 10:0 | FB_PWM_PERIOD | This determines the period of PWM and counted in a MCLK. The 2KHz is recommended, but the value depends on the system condition for externa FB control circuit. |

## FB PWM DUTY

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h1B | 0x0 | x | x | x | FB_PWM_DUTY_WAIT [4:0] | | | | | FB_PWM_DUTY_HIGH [7:0] | | | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 12:8 | FB_PWM_DUTY_WAIT | This determines how many periods of the FAULT_AUTO_READ execution the FB event does not occur, then reduces the active duration time in an arithmetic way. |
| 7:0 | FB_PWM_DUTY_HIGH | This determines the active duration time of the FB_PWM_PERIOD. If the FB event occurs continuously, the active duration will be increased in an arithmetic way every FAULT_AUTO_READ command execution. |

## COMMAND LATENCY

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h1F | 0x1F | x | x | x | x | x | x | x | x | CMD_LATENCY [7:0] | | | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 7:0 | CMD_LATENCY | This determines the interval time between the end of a command and the start of a new command. The value can be increased in the read command execution if the XD12 mclk is so slower than the XC24 mclk, but it is recommended to keep the default value in most cases. |

## DAISIED DEVICE CHANNEL SIZE 1

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h20 | 0x0 | x | DAISIED_DEV_CH_SIZE3 [4:0] | | | | | DAISIED_DEV_CH_SIZE2 [4:0] | | | | | DAISIED_DEV_CH_SIZE1 [4:0] | | | | |

## DAISIED DEVICE CHANNEL SIZE 2

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h21 | 0x0 | x | DAISIED_DEV_CH_SIZE6 [4:0] | | | | | DAISIED_DEV_CH_SIZE5 [4:0] | | | | | DAISIED_DEV_CH_SIZE4 [4:0] | | | | |

## DAISIED DEVICE CHANNEL SIZE 3

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h22 | 0x0 | x | DAISIED_DEV_CH_SIZE9 [4:0] | | | | | DAISIED_DEV_CH_SIZE8 [4:0] | | | | | DAISIED_DEV_CH_SIZE7 [4:0] | | | | |

## DAISIED DEVICE CHANNEL SIZE 4

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h23 | 0x0 | x | DAISIED_DEV_CH_SIZE12 [4:0] | | | | | DAISIED_DEV_CH_SIZE11 [4:0] | | | | | DAISIED_DEV_CH_SIZE10 [4:0] | | | | |

## DAISIED DEVICE CHANNEL SIZE 5

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h24 | 0x0 | x | DAISIED_DEV_CH_SIZE15 [4:0] | | | | | DAISIED_DEV_CH_SIZE14 [4:0] | | | | | DAISIED_DEV_CH_SIZE13 [4:0] | | | | |

## DAISIED DEVICE CHANNEL SIZE 6

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h25 | 0x0 | x | DAISIED_DEV_CH_SIZE18 [4:0] | | | | | DAISIED_DEV_CH_SIZE17 [4:0] | | | | | DAISIED_DEV_CH_SIZE16 [4:0] | | | | |

## DAISIED DEVICE CHANNEL SIZE 7

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h26 | 0x0 | x | DAISIED_DEV_CH_SIZE218 [4:0] | | | | | DAISIED_DEV_CH_SIZE20 [4:0] | | | | | DAISIED_DEV_CH_SIZE19 [4:0] | | | | |

## DAISIED DEVICE CHANNEL SIZE 8

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h27 | 0x0 | x | DAISIED_DEV_CH_SIZE24 [4:0] | | | | | DAISIED_DEV_CH_SIZE23 [4:0] | | | | | DAISIED_DEV_CH_SIZE22 [4:0] | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 14:0 | DAISY_DEV_CH_ SIZEx | This register should be set with a valid number for correct operating. The number means how many channels of the XD12 drivers daisied in the relevant channel are activated. The maximum value of XD12 driver can be up to 12 channels. For future, these registers can support up to 31 channels in incoming XD driver. |

## DAISY SIZE 1

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h30 | 0x0 | x | DAISY_SIZE_CH3 [4:0] | | | | | DAISY_SIZE_CH2 [4:0] | | | | | DAISY_SIZE_CH1 [4:0] | | | | |

## DAISY SIZE 2

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h31 | 0x0 | x | DAISY_SIZE_CH6 [4:0] | | | | | DAISY_SIZE_CH5 [4:0] | | | | | DAISY_SIZE_CH4 [4:0] | | | | |

## DAISY SIZE 3

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h32 | 0x0 | x | DAISY_SIZE_CH9 [4:0] | | | | | DAISY_SIZE_CH8 [4:0] | | | | | DAISY_SIZE_CH7 [4:0] | | | | |

## DAISY SIZE 4

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h33 | 0x0 | x | DAISY_SIZE_CH12 [4:0] | | | | | DAISY_SIZE_CH11 [4:0] | | | | | DAISY_SIZE_CH10 [4:0] | | | | |

## DAISY SIZE 5

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h34 | 0x0 | x | DAISY_SIZE_CH15 [4:0] | | | | | DAISY_SIZE_CH14 [4:0] | | | | | DAISY_SIZE_CH13 [4:0] | | | | |

## DAISY SIZE 6

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h35 | 0x0 | x | DAISY_SIZE_CH18 [4:0] | | | | | DAISY_SIZE_CH17 [4:0] | | | | | DAISY_SIZE_CH16 [4:0] | | | | |

## DAISY SIZE 7

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h36 | 0x0 | x | DAISY_SIZE_CH21 [4:0] | | | | | DAISY_SIZE_CH20 [4:0] | | | | | DAISY_SIZE_CH91 [4:0] | | | | |

## DAISY SIZE 8

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h37 | 0x0 | x | DAISY_SIZE_CH24 [4:0] | | | | | DAISY_SIZE_CH23 [4:0] | | | | | DAISY_SIZE_CH22 [4:0] | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 14:10 | DAISY_SIZE_CHx | This register should be set with a valid number for correct operating. The number means how many XD12 drivers are daisied in the corresponding channel. The maximum 31 XD12 drivers per channel can be daisied. |
| 9:5 | DAISY_SIZE_CHx | The same as above. |
| 4:0 | DAISY_SIZE_CHx | The same as above. |

## BLOCK SIZE 1

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h38 | 0x0 | BLOCK_SIZE_CH2 [7:0] | | | | | | | | BLOCK_SIZE_CH1 [7:0] | | | | | | | |

## BLOCK SIZE 2

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h39 | 0x0 | BLOCK_SIZE_CH4 [7:0] | | | | | | | | BLOCK_SIZE_CH3 [7:0] | | | | | | | |

## BLOCK SIZE 3

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h3A | 0x0 | BLOCK_SIZE_CH6 [7:0] | | | | | | | | BLOCK_SIZE_CH5 [7:0] | | | | | | | |

## BLOCK SIZE 4

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h3B | 0x0 | BLOCK_SIZE_CH8 [7:0] | | | | | | | | BLOCK_SIZE_CH7 [7:0] | | | | | | | |

## BLOCK SIZE 5

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h3C | 0x0 | BLOCK_SIZE_CH10 [7:0] | | | | | | | | BLOCK_SIZE_CH9[7:0] | | | | | | | |

## BLOCK SIZE 6

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h3D | 0x0 | BLOCK_SIZE_CH12 [7:0] | | | | | | | | BLOCK_SIZE_CH11 [7:0] | | | | | | | |

## BLOCK SIZE 7

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h3E | 0x0 | BLOCK_SIZE_CH14 [7:0] | | | | | | | | BLOCK_SIZE_CH13 [7:0] | | | | | | | |

## BLOCK SIZE 8

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h3F | 0x0 | BLOCK_SIZE_CH16 [7:0] | | | | | | | | BLOCK_SIZE_CH15 [7:0] | | | | | | | |

## BLOCK SIZE 9

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h40 | 0x0 | BLOCK_SIZE_CH18 [7:0] | | | | | | | | BLOCK_SIZE_CH17 [7:0] | | | | | | | |

## BLOCK SIZE 10

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h41 | 0x0 | BLOCK_SIZE_CH20 [7:0] | | | | | | | | BLOCK_SIZE_CH19 [7:0] | | | | | | | |

## BLOCK SIZE 11

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h42 | 0x0 | BLOCK_SIZE_CH22 [7:0] | | | | | | | | BLOCK_SIZE_CH21 [7:0] | | | | | | | |

## BLOCK SIZE 12

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ‘h43 | 0x0 | BLOCK_SIZE_CH24 [7:0] | | | | | | | | BLOCK_SIZE_CH23 [7:0] | | | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 15:8 | BLOCK_SIZE_CHx | This register should be set with a valid number for correct operating. The number means how many block zones are daisied in the corresponding channel. The maximum 255 block zones per channel can be daisied. |
| 7:0 | BLOCK_SIZE_CHx | The same as above. |

## CHANNEL ENABLE 1

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 |
|---|---|---|---|---|---|---|---|---|---|
| 'h45 | 0x0 | CH16_EN | CH15_EN | CH14_EN | CH13_EN | CH12_EN | CH11_EN | CH10_EN | CH9_EN |

| Addr | Default | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|
| 'h45 | 0x0 | CH7_EN | CH6_EN | CH5_EN | CH4_EN | CH3_EN | CH2_EN | CH1_EN | CH0_EN |

| Bits | Fields | Descriptions |
|---|---|---|
| 15:0 | CHx_EN | This bit should be set if each corresponding channel is to be activated for the local dimming operation. If any channel is not used, then don't set its bit. |

## CHANNEL ENABLE 2/ SIZE/ LD WIDTH

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 |
|---|---|---|---|---|---|---|---|---|---|
| 'h46 | 0x0 | LD_WIDTH [1:0] | | x | CH_SIZE [4:0] | | | | |

| Addr | Default | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|
| 'h46 | 0x0 | CH24_EN | CH23_EN | CH22_EN | CH21_EN | CH20_EN | CH19_EN | CH18_EN | CH17_EN |

| Bits | Fields | Descriptions |
|---|---|---|
| 15:14 | LD_WIDTH | These bits represent the number of bits contained in one LD data in one block zone.<br>2'b00 : 24 bits/block<br>even if the 24 bits/block are transferred in SPI interface, the valid bits are PAM 8 bits and PWM 14 bits.<br>2'b01 : 24 bits/block, PAM 10 bits + PWM 11 bits.<br>2'b10 : 24 bits/block, PAM 10 bits + PWM 12 bits.<br>2'b11 : 16 bits/block |
| 12:8 | CH_SIZE | These bits represent how many channels are activated. |
| 7:0 | CHx_EN | The same as above CHANNEL ENABLE 1 register. |

## GLOBAL-WRITE DATA

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h60 | 0x0 | x | x | x | x | GLOBAL_WR_DATA [11:0] | | | | | | | | | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 11:0 | GLOBAL_WR_ DATA | This register represents the data to be written to the XD12 drivers daisied. This register should be written with desired value before the global-writing command is executed. |

## GLOBAL FAULT-READ DATA 1

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h61 | 0x0 | FAULT_CH4 [3:0] | | | | FAULT_CH3 [3:0] | | | | FAULT_CH2 [3:0] | | | | FAULT_CH1 [3:0] | | | |

## GLOBAL FAULT-READ DATA 2

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h62 | 0x0 | FAULT_CH8 [3:0] | | | | FAULT_CH7 [3:0] | | | | FAULT_CH6 [3:0] | | | | FAULT_CH5 [3:0] | | | |

## GLOBAL FAULT-READ DATA 3

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h63 | 0x0 | FAULT_CH12 [3:0] | | | | FAULT_CH11 [3:0] | | | | FAULT_CH10 [3:0] | | | | FAULT_CH9 [3:0] | | | |

## GLOBAL FAULT-READ DATA 4

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h64 | 0x0 | FAULT_CH16 [3:0] | | | | FAULT_CH15 [3:0] | | | | FAULT_CH14 [3:0] | | | | FAULT_CH13 [3:0] | | | |

## GLOBAL FAULT-READ DATA 5

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h65 | 0x0 | 0x0 | | | | FAULT_CH19 [3:0] | | | | FAULT_CH18 [3:0] | | | | FAULT_CH17 [3:0] | | | |

## GLOBAL FAULT-READ DATA 6

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h66 | 0x0 | FAULT_CH24 [3:0] | | | | FAULT_CH23 [3:0] | | | | FAULT_CH22 [3:0] | | | | FAULT_CH21 [3:0] | | | |

| Bits | Fields | Descriptions |
|---|---|---|
| 15:12<br>11:8<br>7:4<br>3:0 | FAULT_CHx | These bits represent the fault status from all XD12 drivers daisied in each channel.<br>FAULT_CHx [3] : The thermal status, all bits from each channel are OR-ed.<br>FAULT_CHx [2] : The short status, all bits from each channel are OR-ed.<br>FAULT_CHx [1] : The open status, all bits from each channel are OR-ed.<br>FAULT_CHx [0] : The fb status, all bits from each channel are OR-ed.<br><br>The FAULT_CHx [3:1] are auto-cleared after reading the INTERRUPT register, but the FAULT_CHx [0] remains at the level even if the INTERRUPT register is read-out. And the FAULT_CHx [0] value is driven directly to the FB pin. |

## LDO

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h67 | 0x0008 | x | x | x | x | x | x | x | x | x | x | x | x | LDO [3:0] | | | |

## PORT1 LOCAL-RW DATA 1 ~ PORT8 LOCAL RW DATA 1

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h70~'h77 | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 1 [11:0] ~ PORT8_LOCAL_RW_DATA 1 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 2 ~ PORT8 LOCAL RW DATA 2

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h78~'h7F | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 2 [11:0] ~ PORT8_LOCAL_RW_DATA 2 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 3 ~ PORT8 LOCAL RW DATA 3

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h80~'h87 | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 3 [11:0] ~ PORT8_LOCAL_RW_DATA 3 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 4 ~ PORT8 LOCAL RW DATA 4

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h88~'h8F | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 4 [11:0] ~ PORT8_LOCAL_RW_DATA 4 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 5 ~ PORT8 LOCAL RW DATA 5

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h90~'h97 | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 5 [11:0] ~ PORT8_LOCAL_RW_DATA 5 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 6 ~ PORT8 LOCAL RW DATA 6

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'h98~'h9F | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 6 [11:0] ~ PORT8_LOCAL_RW_DATA 6 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 7 ~ PORT8 LOCAL RW DATA 7

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'hA0~'hA7 | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 7 [11:0] ~ PORT8_LOCAL_RW_DATA 7 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 8 ~ PORT8 LOCAL RW DATA 8

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'hA8~'hAF | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 8 [11:0] ~ PORT8_LOCAL_RW_DATA 8 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 9 ~ PORT8 LOCAL RW DATA 9

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'hB0~'hB7 | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 9 [11:0] ~ PORT8_LOCAL_RW_DATA 9 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 10 ~ PORT8 LOCAL RW DATA 10

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'hB8~'hBF | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 10 [11:0] ~ PORT8_LOCAL_RW_DATA 10 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 11 ~ PORT8 LOCAL RW DATA 11

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'hC0~'hC7 | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 11 [11:0] ~ PORT8_LOCAL_RW_DATA 11 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 12 ~ PORT8 LOCAL RW DATA 12

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'hC8~'hCF | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 12 [11:0] ~ PORT8_LOCAL_RW_DATA 12 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 13 ~ PORT8 LOCAL RW DATA 13

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'hD0~'hD7 | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 13 [11:0] ~ PORT8_LOCAL_RW_DATA 13 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 14 ~ PORT8 LOCAL RW DATA 14

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'hD8~'hDF | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 14 [11:0] ~ PORT8_LOCAL_RW_DATA 14 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 15 ~ PORT8 LOCAL RW DATA 15

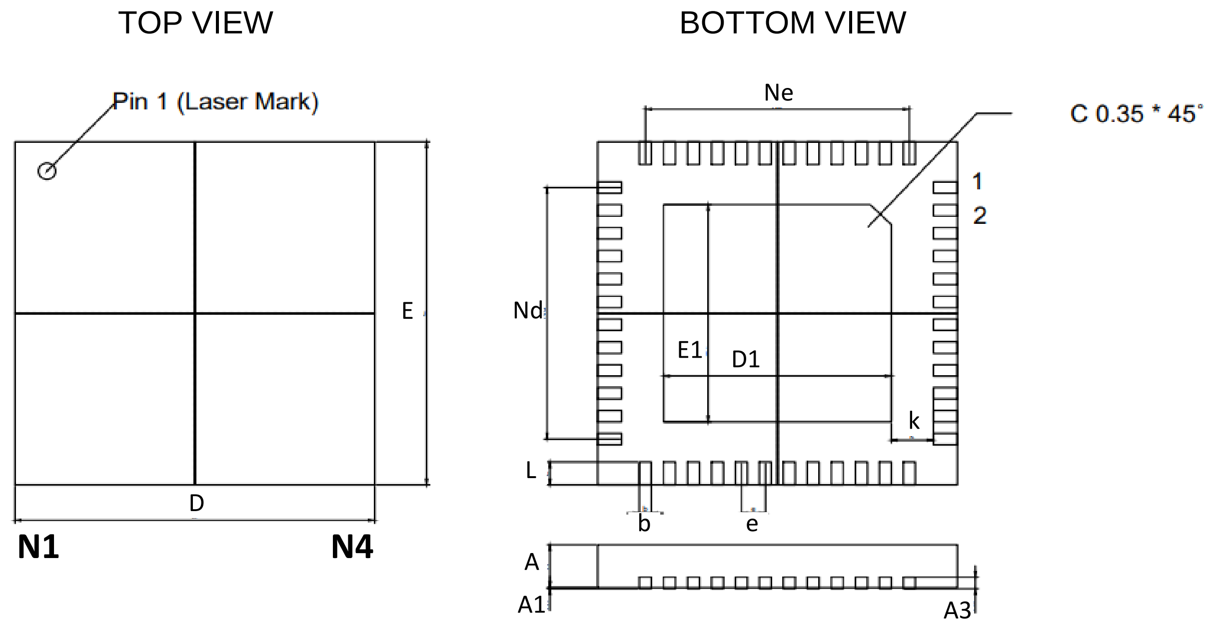
| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'hE0~'hE7 | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 15 [11:0] ~ PORT8_LOCAL_RW_DATA 15 [11:0] | | | | | | | | | | | |

## PORT1 LOCAL-RW DATA 16 ~ PORT8 LOCAL RW DATA 16

| Addr | Default | B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'hE8~'hEF | 0x0 | x | x | x | x | PORT1_LOCAL_RW_DATA 16 [11:0] ~ PORT8_LOCAL_RW_DATA 16[11:0] | | | | | | | | | | | |

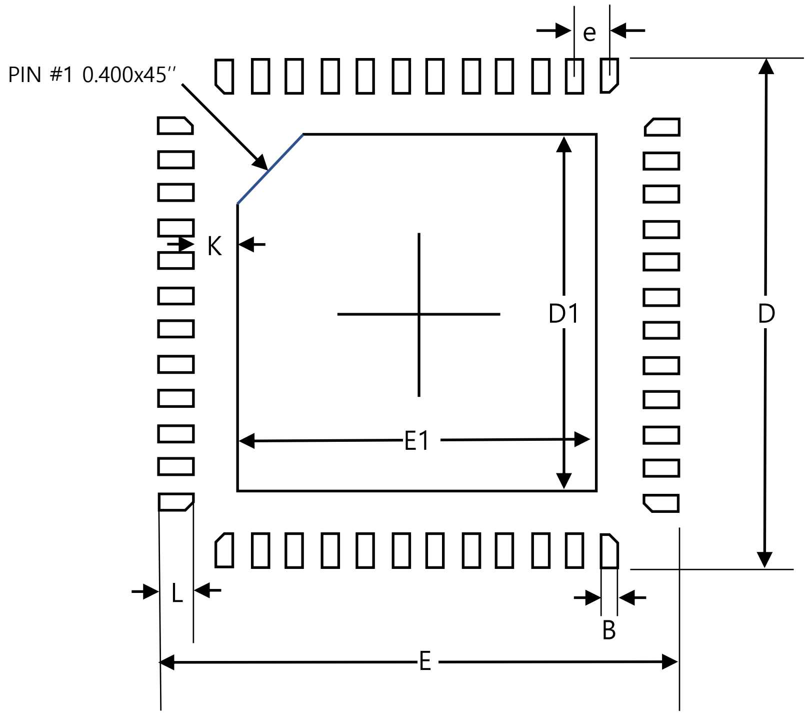
| Bits | Fields | Descriptions |
|---|---|---|
| 11:0 | $PORT_X$_LOCAL_RW_$DATA_Y$ | These registers are used to contain the data to be written/read to/from the XD12 drivers daisied to each channel for accessing the XD12 drivers' registers. These registers should be written before executing the relevant commands, the local-reading or local-writing command.<br><br>It needs to be careful to use that these registers are differently mapped according to the CH_SEG [1:0] of the LOCAL WRITE/READ COMMAND registers<br><br>CH_SEG = 2'b00 : PORT1 to PORT8 are mapped to the channel 1 to 8 in order.<br>CH_SEG = 2'b01 : PORT1 to PORT8 are mapped to the channel 9 to 16 in order.<br>CH_SEG = 2'b10 : PORT1 to PORT8 are mapped to the channel 17 to 24 in order.<br><br>The data order to be transferred to the XD12 drivers daisied in a $channel_X$<br>: $PORT_X$_LOCAL_RW_$DATA_1$ to the 1st driver,<br>$PORT_X$_LOCAL_RW_$DATA_2$ to the 2nd driver, …..<br><br>The data order to be received from the XD12 drivers daisied in a $channel_X$<br>: $PORT_X$_LOCAL_RW_$DATA_1$ from the last driver,<br>$PORT_X$_LOCAL_RW_$DATA_2$ from the (last -1) driver, …..<br><br>Refer to the Figure 10. in detail. |

## Package specification




| Symbol | Dimensions in mm | | |
|---|---|---|---|
| | Min | Typ | Max |
| A | 0.700 | 0.750 | 0.800 |
| A1 | 0.000 | 0.020 | 0.050 |
| A3 | 0.203 REF | | |
| D | 5.900 | 6.000 | 6.100 |
| E | 5.900 | 6.000 | 6.100 |
| D1 | 4.3 | 4.4 | 4.5 |
| E1 | 4.3 | 4.4 | 4.5 |
| k | 0.250 REF | | |
| b | 0.140 | 0.200 | 0.250 |
| e | 0.400 BSC | | |
| Nd | 4.400 BSC | | |
| Ne | 4.400 BSC | | |
| L | 0.350 | 0.400 | 0.450 |

## Footprint




| Symbol | Dimensions in mm | | |
|---|---|---|---|
| | Min | Nom | Max |
| A | 0.7 | 0.75 | 0.8 |
| A1 | 0.00 | 0.02 | 0.05 |
| A2 | 0.203 REF | | |
| B | 0.15 | 0.2 | 0.25 |
| D | 5.9 | 6 | 6.1 |
| E | 5.9 | 6 | 6.1 |
| D1 | 4.6 | 4.7 | 4.8 |
| E1 | 4.6 | 4.7 | 4.8 |
| e | 0.4 BSC | | |
| K | 0.25 REF | | |
| L | 0.3 | 0.4 | 0.5 |